CS774: Spectral Graph Theory Notes

Om Joglekar

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1 Random Walks on Graphs

1.1 Introduction

Spectral graph theory is a subject lying in the intersection of linear algebra and graph theory. To feel ones way around a graph, one can employ certain algorithms - the popular ones being BFS and DFS. There is yet another contender known as random walks.

Random walks are one of the important contact points between the subjects of linear algebra and graph theory. Below, we define a random walk and also some associated vectors and matrices.

We will work with finite and undirected graphs unless explicitly stated. In most cases, we will work with possibly non-simple graphs and we will state whether the graph is simple if required. Furthermore, all vectors will be treated as row vectors since this is the classical convention used when the theory was developed. This also ties up well with coding theory.

Definition 1.1. Let $G = (V, E)$ be a connected graph. A ***random walk*** is a randomised process that takes place on the graph as follows. We begin with a fixed vertex $v_0 \in V$ and having chosen vertices $v_0, v_1, \ldots, v_t$ for $t \geq 0$, the next vertex v_{t+1} is chosen to be a uniformly random neighbour of x_t.

We employ the following notation.

Notation 1.2. In a graph $G = (V, E)$, the set of neighbours of a vertex $v \in V$ will be denoted as $N(v)$ and the choice of a neighbour w uniformly from this set will be denoted as

$$w \sim \mathrm{Unif}(N(v))$$

So far, it is clear that this is a combinatorial viewpoint of random walks. We now go into the linear algebraic details.

Definition 1.3. In a graph $G = (V, E)$, the ***transition probability vector*** or the ***state vector*** of a vertex $v \in V$ is defined to as follows. $p_v \in \mathbb{R}^n$ (where $|V| = n$) is the probability distribution vector of vertex v whose entry at vertex $u \in V$ is given by

$$(p_v)_u = \begin{cases} \frac{1}{d(u)} & \text{if } \{u, v\} \in E \\ 0 & \text{otherwise} \end{cases}$$

where $d(u)$ is the degree of the vertex u.

The example below makes this definition clear. For the graph in figure 1 the transition probability vectors at the vertices v_7 and v_2 are given by

$$p_{v_7} = \begin{bmatrix} \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & 0 \end{bmatrix} \in \mathbb{R}^n$$
$$p_{v_2} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^n$$

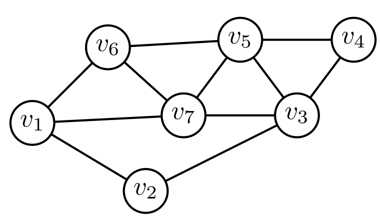

Figure 1: A generic arbitrary graph

We also define an associated random walk matrix below.

Definition 1.4. Given a graph G on n vertices, the ***random walk matrix***, usually denoted M is an $n \times n$ real matrix whose row are indexed by vertices of the graph and the row corresponding to a vertex is given by the transition probability vector of that vertex.

For the graph from figure 1 the associated random walk matrix is given by

$$M = \begin{bmatrix} 0 & \frac{1}{3} & 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & 0 & 0 & 0 & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{4} & 0 & \frac{1}{4} & 0 & \frac{1}{4} & \frac{1}{4} & 0 \end{bmatrix}$$

One observes that if we look at the zero and non-zero entries of this matrix, then it is very close to the adjacency matrix of the graph. One can say that the rows of the adjacency matrix have been scaled. The proposition below formalises this observation.

Proposition 1.5. *The random walk matrix of a graph with adjacency matrix A and degree matrix D (diagonal matrix with entries being degrees of vertices) is given by*

$$M = D^{-1}A$$

Proof. The proof is trivial and follows by comparing entries on both sides of the equation. ∎

Some important properties of the rows of this matrix, that is, the transition probability vectors, are that all the entries are non-negative and all the entries sum up to 1 since we are adding the probabilities of all possible outcomes. This leads us to the following result.

Proposition 1.6. *Let M be a random walk matrix of some graph. Let $v \in \mathbb{R}^n$ be a vector with non-negative entries which add up to* 1*. Then the vector given by*

$$w = vM$$

is also a non-negative vector whose entries add up to 1.

Proof. Since all entries of M and v are non-negative, the same can be said about w. To verify the claim of entries adding up to 1, we take the dot product with the all 1's vector $j \in \mathbb{R}^n$ and check if the result is 1 or not.

$$wj^\top = vMj^\top = vj^\top = 1$$

Note that we have used $Mj^\top = j^\top$. This follows because every row of M adds up to 1. ∎

One notes that we now need to consider connected graphs so that we can avoid situations where it not possible to traverse between vertices of a graph. Furthermore, if a vertex is on its own, then the probability vector would be the zero vector which is not something we would like to deal with.

This simplification is justified. Using the theory we shall develop for connected graphs, we can apply it to each component. Furthermore, the matrix will be block diagonal and will behave 'nicely'.

Now a natural question to ask is what happens if we keep applying this over and over. Does a limiting vector exist? For what class of matrices can we do this?

Now firstly it is clear that we want the eigenvalues to have length (they could be complex) between 0 and 1 else they blow up when we consider powers of the matrix.

To avoid complex eigenvalues, we restrict our attention to matrices yielding real eigenvalues only. We further simplify our lives by considering only real symmetric matrices so that all the eigenvalues are real by the spectral theorem.

Notation 1.7. We denote the set of all $n \times n$ real symmetric matrices by $\mathrm{Sym}(M_n(\mathbb{R}))$.

Proposition 1.8. *The random walk matrix of a graph is symmetric if and only if the graph is regular.*

Proof. If the graph is regular, then the degree matrix D becomes a scalar multiple of the identity matrix and hence $D^{-1}A$ will be symmetric.

Conversely, suppose $D^{-1}A$ is symmetric. Then for distinct i and j between 1 and n, we have

$$\frac{A_{i,j}}{d_i} = \frac{A_{j,i}}{d_j}$$

Since A is symmetric, we must have $d_i = d_j$ which gives that all the degrees must be equal. Thus D must be a scalar multiple of the identity. ■

The natural next question to ask is about the eigenvalues of such a real symmetric matrix M of non-negative entries whose rows add up to 1. The following result will answer this.

Proposition 1.9. *Let A be the adjacency matrix of a d–regular graph on n vertices. Then d is the largest eigenvalue of A by modulus, is attained at the eigenvector $j = (1, 1, \ldots, 1) \in \mathbb{R}^n$ and the algebraic multiplicity of d gives the number of connected components of the graph.*

Proof. It is clear that $Aj^\top = dj^\top$ or equivalently $jA = dj$. Indeed, recall that left and right eigenvalues of a matrix are the same because A and $A^\top$ have the same eigenvalues and are in fact similar through an invertible matrix P over the same field.

To show that d is indeed the largest eigenvalue, one may directly make use of the Gershgorin circle theorem and the claim is proved. We however highlight another way through the Laplacian.

Recall that the Laplacian matrix is given by $L = D - A$. For our d–regular graph, it takes the form $L = dI - A$. We also make use of the sign free Laplacian matrix $Q = D + A$ which in our case is $dI + A$. We prove a general lemma.

Lemma The Laplacian and sign free Laplacian matrices of any graph are positive semi-definite.

Proof of Lemma. One way is to define the $|V| \times |E|$ incidence matrix M of a graph which has entry 1 in the (v, e)–th position if the edge e is incident at vertex v and 0 elsewhere. One also defines the directed incidence matrix for a directed graph of the same size as before whose (v, e)–th position is 1 if v is the tail of e, -1 if it is the head and 0 otherwise.

Now observe that $Q = MM^\top$ and after assigning the edges of our undirected graph any arbitrary orientation, $L = NN^\top$. This directly shows that L and Q are PSD since any matrix of the form $XX^\top$ is PSD.

Alternatively, consider any edge $\{i, j\} \in E$. Now define a smaller mini-Laplacian matrix $L_{i,j}$ whose (u, v)–th entry given by

$$(L_{i,j})_{(u,v)} = \begin{cases} 1 & \text{if } u = i, v = i \\ -1 & \text{if } u = i, v = j \\ -1 & \text{if } u = j, v = i \\ 1 & \text{if } u = j, v = j \\ 0 & \text{otherwise} \end{cases}$$

Note that this if we delete all the 0 entries, this is simply the Laplacian of the edge graph with two vertices and one edge between them. Also note that $L_{i,j}$ is clearly PSD. Now one notes by comparing entries on both sides that

$$L = \sum_{\{i,j\} \in E} L_{i,j}$$

Since the sum of PSD matrices is PSD, we are done. Explicitly, note that $xL_{i,j}x^\top = (x_i - x_j)^2$.

In fact without using these smaller Laplacians one may perform algebraic manipulations to directly obtain

$$xLx^\top = xDx^\top - xAx^\top = \sum_{\{i,j\}\in E} (x_i - x_j)^2 \geq 0$$

showing that L is PSD. A similar analysis can be done for Q. \\\

Recall that PSD matrices have only non-negative eigenvalues. Since the eigenvalues of $dI - A$ are of the form $d - \lambda$ where λ varies over eigenvalues of A, we have $\lambda \leq d$. Similarly from the fact that Q is PSD we obtain $\lambda \geq -d$. This gives

$$|\lambda| \leq d$$

where λ varies over all eigenvalues of A. This proves that d is indeed the largest eigenvalue by modulus.

As for the result for connected components, we again prove a general lemma from which the result follows.

Lemma The number of connected components of any graph G is the multiplicity of 0 *as an eigenvalue of the Laplacian matrix L, that is the nullity of L.*

Proof of Lemma. Suppose G is connected. The kernel of L is given by

$$\ker(L) = \left\{v \in \mathbb{R}^n \mid Lv^\top = 0\right\}$$

Note that the left and right kernels are same because L is symmetric.

Suppose $v \in \ker(L)$. Then

$$Lv^\top = 0 \implies vLv^\top = 0 \implies \sum_{\{i,j\}\in E} (v_i - v_j)^2 = 0$$

Thus all the components v_i of the vector v are equal because the graph is connected. Conversely any vector which is a scalar multiple of the all 1's vector j lies in the kernel. This shows that the kernel is one dimensional. Thus the multiplicity of 0 as an eigenvalue is the nullity which is 1.

Now suppose G can be broken down into its component graphs as

$$G = G_1 \sqcup \cdots \sqcup G_k$$

Ordering the vertices according to the components, we can block diagonalise the Laplacian matrix to have blocks $L_1, \ldots, L_k$ of sizes $n_1, \ldots, n_k$ where n_i is the number of vertices in G_i.

Break a vector $v \in \mathbb{R}^n$ into $v_1, v_2, \ldots, v_k$ where v_1 has the first n_1 coordinates and so on. Then $v \in \ker(L)$ implies that $v_i \in \ker(L_i)$ for every $i = 1, 2, \ldots, k$ which by the previous case for $k = 1$ gives that every component of each v_i must be equal.

Conversely, any vector with such a condition clearly lies in the kernel.

Thus every vector $v \in \ker(L)$ has the first n_1 components equal to some μ_1, the next n_2 components equal to some μ_2 and so on. This clearly shows that the dimension of the kernel is k.

More precisely, one may write

$$\ker(L) = \ker(L_1) \oplus \cdots \oplus \ker(L_k)$$

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Corollary 1.10. *Random walk matrices of connected regular graphs will always have all eigenvalues real and between* -1 *and* 1.

1.2 Lazy Random Walks

Now having settled the issue of the class of matrices we now ask if for such matrices, will a limiting vector be obtained. Unfortunately, we need more conditions and restrictions as is shown by the following simplest example.

Consider K_2, that is the edge graph with 2 vertices and an edge between them. The adjacency matrix, which is the same as the random walk matrix is given by

$$A = M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The eigenvalues are 1 and -1 and the limiting behaviour is bad because $A^2 = I$ which tells us that the powers of A alternate between A and I.

Such an oscillation will also be observed, in general, in all bipartite graphs since we will just end up oscillating between the two colours and never stabilise.

We note that 1 will always be an eigenvalue of all random walk matrices. The same cannot be said about the eigenvalue -1. From the example we just saw, we first try to get rid of the eigenvalues -1. This is done by adding the identity matrix. However, now the eigenvalues are between 0 and 2. Thus we divide by 2.

Definition 1.11. Given a random walk matrix M of a graph, we define the ***lazy random walk matrix*** W to be

$$W = \frac{I + M}{2}$$

In the graph setting, we simply added the degree of the vertex many more loops at a given vertex and normalised so that the row sums are back to 1. Now if we begin at a vertex, we first have a $50 - 50$ chance of moving or staying fixed. If we decide to move, we move to a vertex uniformly chosen from the neighbours of the current vertex as before. This motivates the terminology 'lazy' since half the time we stay at the given vertex without moving, according to M, of course.

Now we analyse the limiting behaviour of the vector when the matrix W keeps acting on it.

Proposition 1.12. *Let $P_0 \in \mathbb{R}^n$ be the distribution vector at some vertex of a connected regular graph G on n vertices. Consider the lazy random walk matrix W. We have*

$$\lim_{m \to \infty} P_0 W^m = \frac{1}{n} j$$

where $j \in \mathbb{R}^n$ is the all 1's vector.

Proof. Suppose the eigenvalues of W are $1 = \lambda_1 \geq \cdots \geq \lambda_n \geq 0$. We shall use the result that

$$\lambda_1 - \lambda_2 \geq \frac{1}{n^2}$$

This result will be proved later.

We want to show the following (equivalent to the given theorem)

$$\lim_{m \to \infty} \left\| P_0 W^m - \frac{1}{n} j \right\| = 0 \quad \text{for every } P_0 \in \mathbb{R}^n$$

This can be reduced to showing the result for $P_0 = e_i$, where $e_i \in \mathbb{R}^n$ for $i = 1, 2, \ldots, n$ are the standard basis vectors of $\mathbb{R}^n$, because of linearity and the triangle inequality.

Fix some $i \in [n]$. We want to show the following

$$L = \lim_{m \to \infty} \| n e_i W^m - j \| = 0$$

Pick vectors $u_1, \ldots, u_n$ which form an unit eigenbasis of W with eigenvalues $\lambda_1, \ldots, \lambda_n$ respectively. We know that $\lambda_1 = 1$ and $u_1 = (1/\sqrt{n})j$.

By the spectral theorem we have

$$W = \lambda_1 u_1^t u_1 + \cdots + \lambda_n u_n^t u_n$$

Thus in general we have by orthogonality, that is, $u_i u_j^t = 0$ for $i \neq j$:

$$W^m = \lambda_1^m u_1^t u_1 + \cdots + \lambda_n^m u_n^t u_n$$

Thus the limit becomes

$$L = \lim_{m\to\infty} \left\| n e_i \left(\lambda_1^m u_1^t u_1 + \cdots + \lambda_n^m u_n^t u_n \right) - j \right\|$$

Note that $e_i u_j^t = (u_j)_i$, that is, the i–th component of u_j for every $j = 1, 2, \ldots, n$. Further note that

$$n\lambda_1^m (u_1)_i u_1 = n \frac{1}{\sqrt{n}} \frac{1}{\sqrt{n}} j = j$$

Thus we have

$$\begin{aligned} L &= \lim_{m\to\infty} \|n\lambda_2^m (u_2)_i u_2 + \cdots + n\lambda_n^m (u_n)_i u_n\| \\ &\leq \lim_{m\to\infty} n \|\lambda_2^m (u_2)_i u_2\| + \cdots + n \|\lambda_n^m (u_n)_i u_n\| \\ &\leq \lim_{m\to\infty} n \left(|\lambda_2^m| + \cdots + |\lambda_n^m|\right) \\ &\leq \lim_{m\to\infty} n(n-1) |\lambda_2|^m \\ &= \lim_{m\to\infty} n(n-1)\lambda_2^m \\ &\leq \lim_{m\to\infty} n(n-1) \left(1 - \frac{1}{n^2}\right)^m \\ &= 0 \end{aligned}$$

Note that the second last follows from the result on difference of eigenvalues used earlier. As remarked, we shall prove this later. ■

We make a few remarks on this theorem. One observes that no matter what the graph is or which vertex we start form, after an infinite amount of time, the probability of being at a given vertex is independent of the choice of it since it gets equally distributed among all vertices.

Further, suppose we let $1 - \lambda_2 = \varepsilon$ then we derive

$$\left\| e_i W^m - \frac{1}{n} j \right\| \leq (1 - \varepsilon)^t n$$

Thus if we want the error to be smaller than some constant δ, say, we solve

$$(1 - \varepsilon)^t n \leq \delta \implies t \geq \frac{1}{\varepsilon} \left(\ln(n) - \ln(\delta)\right)$$

Thus, we have $t = O\left(\dfrac{\ln(n)}{\varepsilon}\right)$.

Definition 1.13. A probability distribution vector $p \in \mathbb{R}^n$ is said to be a ***stationary distribution vector*** with respect to a matrix W if there exists some $\ell \in \mathbb{N}$ such that

$$pW^\ell = pW^{\ell+1}$$

Note that the linear combination of two such distributions is again a stationary distribution.

We have a theorem without proof. We may prove it in subsequent lectures if time permits.

Theorem 1.14. *Let G be any connected graph and let W be its associated lazy random walk matrix. Then W admits a unique stationary distribution vector (of course, up to scaling) given by the vector $\pi \in \mathbb{R}^n$ whose coordinate corresponding to vertex v is given by*

$$\pi_v = \frac{\deg(v)}{2\,|E|}$$

Note that when G is regular, the above theorem recovers the vector $(1/n)j$ we had in proposition 1.12.

We shall now go over some advanced linear algebra theorems which shall prove very useful to us.

1.3 Advanced Linear Algebra

Theorem 1.15 (Courant-Fischer theorem)**.** *Let M be any complex Hermitian matrix with (real) eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$. Suppose $S \subseteq \mathbb{C}^n$ denotes a subspace. Then we have for every $k = 1, 2, \ldots, n$,*

$$\lambda_k = \min_{\substack{S \subseteq \mathbb{C}^n \\ \dim(S)=k}} \max_{x \in S\setminus\{0\}} \frac{xMx^*}{xx^*}$$

Proof. Pick an orthonormal eigenbasis $\{v_1, \ldots, v_n\}$ of M with eigenvalues $\{\lambda_1, \ldots, \lambda_n\}$. Such an eigenbasis exists because of the spectral theorem for Hermitian matrices.

Set $S = \operatorname{span}\{v_1, \ldots, v_k\}$ to be a subspace of dimension k. Pick any non-zero $x \in S$ and write it as

$$x = c_1 v_1 + \cdots + c_k v_k$$

Then we have

$$\begin{aligned}
\frac{xMx^*}{xx^*} &= \frac{(c_1 v_1 + \cdots + c_k v_k)\, M\, (\overline{c_1} v_1^* + \cdots + \overline{c_k} v_k^*)}{|c_1|^2 + \cdots + |c_k|^2} \\
&= \frac{(c_1 \lambda_1 v_1 + \cdots + c_k \lambda_k v_k)\, (\overline{c_1} v_1^* + \cdots + \overline{c_k} v_k^*)}{|c_1|^2 + \cdots + |c_k|^2} \\
&= \frac{\lambda_1 |c_1|^2 + \cdots + \lambda_k |c_k|^2}{|c_1|^2 + \cdots + |c_k|^2} \\
&\leq \lambda_k
\end{aligned}$$

We have shown that for a particular choice of S, any non-zero vectors makes the fraction smaller than λ_k. This guarantees that minimising over all possible S and then picking an optimum x in it, will also keep the result smaller than λ_k. Thus, now we show that λ_k is indeed the minimum by picking an arbitrary subspace S and obtaining some vector in it which makes the fraction at least λ_k.

To this extent, let $S \subseteq \mathbb{C}^n$ be an arbitrary subspace of dimension k. Pick $W = \operatorname{span}\{v_1, \ldots, v_{k-1}\}$. Then the dimension of $W^\perp$ is $n - k + 1$ while that of S is k. Thus the intersection will have dimension

$$\dim(W^\perp \cap S) = n - k + 1 + k - \dim(W^\perp + S) = n + 1 - \dim(W^\perp + S) \geq n + 1 - n = 1$$

Thus pick some non-zero $x \in W^\perp \cap S$ to be a vector is S. Since $x \in W^\perp$, it is orthogonal to each one of the vectors $v_1, \ldots, v_{k-1}$. Consequently, x is only a linear combination of the vectors $v_k, \ldots, v_n$. Hence we get

$$\begin{aligned}
\frac{xMx^*}{xx^*} &= \frac{(c_k v_k + \cdots + c_n v_n)\, M\, (\overline{c_k} v_k^* + \cdots + \overline{c_n} v_n^*)}{|c_k|^2 + \cdots + |c_n|^2} \\
&= \frac{(c_k \lambda_k v_k + \cdots + c_n \lambda_n v_n)\, (\overline{c_k} v_k^* + \cdots + \overline{c_n} v_n^*)}{|c_k|^2 + \cdots + |c_n|^2} \\
&= \frac{\lambda_k |c_k|^2 + \cdots + \lambda_n |c_n|^2}{|c_k|^2 + \cdots + |c_n|^2} \\
&\geq \lambda_k
\end{aligned}$$

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Below in an important corollary often referred to as the Rayleigh quotienting method.

Corollary 1.16 (Rayleigh quotients)**.** *Let M be any complex Hermitian matrix with (real) eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$. Then if $R(x)$ denotes the fraction considered previously, we have*

$$\begin{array}{rcll}
\lambda_n & = & \max\limits_{x\neq 0} & R(x) \\
\lambda_{n-1} & = & \max\limits_{\substack{x\perp v_n \\ x\neq 0}} & R(x) \\
 & \vdots & & \\
\lambda_{n-i} & = & \max\limits_{\substack{x\perp span\{v_n,\ldots,v_{n-i+1}\} \\ x\neq 0}} & R(x) \\
 & \vdots & & \\
\lambda_1 & = & \max\limits_{\substack{x\perp span\{v_n,\ldots,v_2\} \\ x\neq 0}} & R(x)
\end{array}$$

Proof. Let $k = n-i$. The condition that $x \perp \text{span}\{v_{k+1},\ldots,v_n\}$ implies that x lies entirely in the subspace $W = \text{span}\{v_1,\ldots,v_k\}$.

Any non-zero vector $x \in W$ can be written as $x = \sum_{j=1}^{k} c_j v_j$. Since the eigenvalues are ordered $\lambda_1 \leq \cdots \leq \lambda_k$, we can bound the Rayleigh quotient:

$$R(x) = \frac{\sum_{j=1}^{k} \lambda_j |c_j|^2}{\sum_{j=1}^{k} |c_j|^2} \leq \frac{\lambda_k \sum_{j=1}^{k} |c_j|^2}{\sum_{j=1}^{k} |c_j|^2} = \lambda_k$$

Thus, the maximum value over this set is at most λ_k. We satisfy this upper bound by choosing the specific vector $x = v_k$. Since $v_k \in W$ and $R(v_k) = \lambda_k$, the maximum is exactly attained at λ_k. ■

We also have the dual versions of the Courant-Fischer and the Rayleigh quotients as follows:

$$\lambda_k = \max_{\substack{S\subseteq\mathbb{C}^n \\ \dim(S)=n-k+1}} \min_{x\in S\setminus\{0\}} \frac{xMx^*}{xx^*}$$

and

$$\lambda_i = \min_{\substack{x\perp\text{span}\{v_1,\ldots,v_{i-1}\} \\ x\neq 0}} R(x)$$

1.4 The Cheeger Constant

We now begin our discussion on the Cheeger constant and give some fundamental inequalities related to it.

Definition 1.17. Given an undirected graph G we define the ***edge isoperimetric parameter*** to be

$$\Phi_E(k) = \min_{S\subseteq V, |S|=k} |\partial S|$$

where ∂S is the set of all edges going between the set S and the set S^c so that $|\partial S|$ is the number of edges leaving the set S.

Definition 1.18. Given an undirected graph G we define the ***Cheeger constant*** to be

$$h(G) = \min\left\{\frac{|\partial A|}{|A|} : A \subseteq V, 0 < |A| \leq \frac{|V|}{2}\right\}$$

It can be written for n vertices as

$$h(G) = \min_{1 \le k \le n/2} \frac{\Phi_E(k)}{k}$$

This constant is clearly 0 for a disconnected graph. Thus the only interesting cases are when the graph is connected. This constant is closely related to the second largest eigenvalue of the graph.

The goal of this lecture and the next lecture is to prove the following theorem.

Theorem 1.19 (Cheeger Inequality). *For any undirected graph on n vertices we have*

$$2h(G) \ge \lambda \ge \frac{h(G)^2}{2\Delta(G)}$$

where $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_n \le 2\Delta$ and $\lambda = \lambda_2$. These are the eigenvalues of the Laplacian.

This theorem will be proven as two propositions concerning the two bounds we have. The below proposition 1.20 concerns the upper bound while proposition 1.23 proves the lower bound. Together, they are the same as Theorem 1.19.

Proposition 1.20. *For any undirected graph we have*

$$\lambda \le 2h(G)$$

Proof. From the dual of corollary 1.16 discussed just after the corollary, we know that

$$\lambda = \min_{\substack{x \in \mathbb{R}^n \\ x \perp j}} \frac{xLx^t}{xx^t}$$

Pick S to be any cut of the vertex set with $|S| = a$ and $|S^c| = b = n - a$. Define the vector $x_S \in \mathbb{R}^n$ as

$$x_S(v) = \begin{cases} -b & \text{if } v \in S \\ a & \text{if } v \notin S \end{cases}$$

Then $\langle j, x_S \rangle = jx_S^t = |S|(-b) + |S^c|(a) = 0$. Thus, the constructed vector is orthogonal to the all $1's$ vector j which happens to be v_1 up to a constant. Hence we get

$$\begin{aligned}
\lambda &\le \frac{x_S L x_S^t}{x_S x_S^t} \\
&= \frac{\sum_{\{i,j\} \in E} (x_S(v_i) - x_S(v_j))^2}{ab^2 + ba^2} \\
&= \frac{(a+b)^2 |\partial S|}{ab(a+b)} \\
&= \frac{a+b}{b} \cdot \text{cut}(S) \\
&\le 2 \cdot \text{cut}(S)
\end{aligned}$$

where $\text{cut}(S) = \dfrac{\partial S}{|S|}$. Notice that we have used $\dfrac{a+b}{b} \le \dfrac{b+b}{b} \le 2$.

Thus we have shown that for any cut S with $|S| \le a \le n/2$, we have $\text{cut}(S)$ at least $\lambda/2$. Since the Cheeger constant is simply the minimum over all possible cuts, it will also be greater than $\lambda/2$. ∎

Proving the other inequality is not so straightforward since now we have to exhibit a particular nice cut with the cut parameter $\sqrt{2\lambda\Delta(G)}$.

We develop several lemmas towards this end.

Lemma 1.21. *Let $g \in \mathbb{R}^n$ be a vector orthogonal to j. Without loss of generality by choosing $-g$ we can assume that $|\{v \in V \mid g(v) > 0\}| = |V^+| \leq n/2$. Here, $g(v)$ denotes the v–th component of the vector g. Define the vector $f \in \mathbb{R}^n$ as*

$$f(v) = \begin{cases} g(v) & \text{if } v \in V^+ \\ 0 & \text{otherwise} \end{cases}$$

Then

$$\frac{fLf^t}{ff^t} \leq \lambda$$

for some specific choice of the vector g.

Proof. We have

$$fLf^t = \sum_{v\in V} (fL)(v)f(v) = \sum_{v \in V^+} (fL)(v)g(v)$$

Now for $v \in V^+$ we get

$$\begin{aligned}(fL)(v) &= (fD)(v) - (fA)(v) \\ &= f(v)d(v) - \sum_{w\sim v} f(w) \\ &= g(v)d(v) - \sum_{w\sim v} f(w) \\ &\leq g(v)d(v) - \sum_{w\sim v} g(w) \\ &= (gL)(v)\end{aligned}$$

Here $w \sim v$ means that $w \in V$ is adjacent to v in the graph. Finally we get

$$fLf^t = \sum_{v\in V^+} (fL)(v)g(v) \leq \sum_{v\in V^+} (gL)(v)g(v)$$

Also we have that

$$f^tf = \sum_{v\in V} f(v)^2 = \sum_{v\in V^+} g(v)^2$$

In particular let g be the vector chosen such that $gL = \lambda g$, that is, g optimises the following

$$\lambda = \min_{\substack{x\in\mathbb{R}^n, x\neq 0 \\ x\perp j}} \frac{xLx^t}{xx^t} = \frac{gLg^t}{gg^t}$$

Then we get

$$fLf^t \leq \sum_{v\in V^+} \lambda g(v)^2 = \lambda ff^t$$

This completes the proof of the lemma. ∎

Lemma 1.22. *Let f be chosen as in the above lemma 1.21. We define*

$$B_f = \sum_{\substack{u,v: u\sim v \\ f(u)\geq f(v)}} \left(f(u)^2 - f(v)^2\right)$$

Then we have

$$B_f \leq \sqrt{2\Delta(fLf^t)(ff^t)}$$

Proof. The peculiar condition of $f(v) > f(u)$ serves the purpose of counting each edge only once with a fixed order of its endpoints.

We make use of the identity $(a+b)^2 \leq 2(a^2+b^2)$. We have

$$\begin{aligned} B_f &= \sum_{\substack{u,v:u\sim v\\ f(u)\geq f(v)}} (f(u)+f(v))\,(f(u)-f(v)) \\ &\leq \sqrt{\left(\sum (f(u)+f(v))^2\right)\left(\sum (f(u)-f(v))^2\right)} \\ &\leq \sqrt{\left(\sum 2(f(u)^2+f(v)^2)\right)(fLf^t)} \\ &\leq \sqrt{2\Delta(ff^t)(fLf^t)} \end{aligned}$$

In the first step we used the Cauchy-Schwarz inequality. In the next step, for the first term we used the identity stated before and the second term was exactly equal to fLf^t.

In the last step, the sum is over all edges $\{u,v\}$ and for each such edge, the summand is $f(u)^2+f(v)^2$. We might as well sum over each vertex u in which case the summand is $d(u)f(u)^2$. Each degree is bounded above by the maximum degree Δ which then justifies the last step. ∎

Proposition 1.23. *Borrowing notation from lemma 1.22 we have*

$$h(G)(ff^t) \leq B_f$$

Consequently we have

$$\frac{h(G)^2}{2\Delta(G)} \leq \lambda$$

Proof. Suppose f takes the distinct values $\alpha_0 = 0 < \alpha_1 < \cdots < \alpha_r$ on the vertices of G. For each $0 \leq i \leq r$ we define

$$L_i = \{v \in V(G) \mid f(v) \geq \alpha_i\}$$

It follows by definition that

$$L_r \subseteq L_{r-1} \subseteq \cdots \subseteq L_1 \subseteq L_0 = V(G)$$

Suppose $e = \{u,v\}$ is an edge. Without loss of generality assume that $f(u) > f(v)$. Edges with both vertices having the same value of f do not contribute to B_f. Suppose $f(u) = \alpha_j$ and $f(v) = \alpha_i$ with $j > i$ by assumption. Then the edge contributes $\alpha_j^2 - \alpha_i^2$ ot the sum B_f.

This contribution can also be written be written as

$$\alpha_j^2 - \alpha_i^2 = \sum_{t=i+1}^{j} (\alpha_t^2 - \alpha_{t-1}^2)$$

Hence our sum becomes

$$B_f = \sum_{\substack{u\sim v\\ f(u)>f(v)}} \sum_{t=i+1}^{j} (\alpha_t^2 - \alpha_{t-1}^2)$$

For each edge $\{u,v\}$, the term $\alpha_t^2 - \alpha_{t-1}^2$ is added if and only if $t \in (i,j]$, that is, $f(v) < \alpha_t \leq f(u)$. This means that $u \in L_t$ but $v \notin L_t$ and hence this edge contributes a factor of $\alpha_t^2 - \alpha_{t-1}^2$ if and only if $\{u,v\} \in \partial L_t$. This allows us to express B_f as

$$B_f = \sum_{t=1}^{r} \left(\alpha_t^2 - \alpha_{t-1}^2\right) |\partial L_t|$$

Now by definition of $h(G)$, since $|L_t| \leq \frac{n}{2}$ for each $t = 1, 2, \ldots, r$ (support of f is at most $n/2$), we have

$$h(G) \leq \frac{|\partial L_t|}{|L_t|}$$

Using this we get

$$
\begin{aligned}
B_f &= \sum_{t=1}^{r} \left(\alpha_t^2 - \alpha_{t-1}^2\right) |\partial L_t| \\
&\geq h(G) \sum_{t=1}^{r} \left(\alpha_t^2 - \alpha_{t-1}^2\right) |L_t| \\
&= h(G) \left(\alpha_0^2 \, |L_1| + \alpha_r^2 \, |L_r| + \sum_{t=1}^{r-1} \alpha_t^2 (|L_t| - |L_{t+1}|) \right) \\
&= h(G) \left(\alpha_r^2 \, |L_r| + \sum_{t=1}^{r-1} \alpha_t^2 \, |V_t| \right) \\
&= h(G) \sum_{t=1}^{r} \alpha_t^2 \, |V_t|
\end{aligned}
$$

Here V_t is the set $L_t \setminus L_{t-1}$ which is the set of those vertices with f–value exactly α_t. Note that $V_r = L_r$ since there are no vertices with value greater than α_r and $V_0 = V(G)$.

In the third step above, we have simply written the sum in a different form by gathering the coefficients of the α_t rather than the previous form which factors out the $|L_t|$. In the subsequent fourth step, we simply observed that $\alpha_0 = 0$.

Now we might as well start the summation from $t = 0$ since $\alpha_0 = 0$. Thus, for each given t, we count the number of vertices with the given f–value α_t and add α_t^2. This is the same as summing over each vertex and adding the square of its f–value. Thus,

$$
B_f \geq h(G) \sum_{t=0}^{r} \alpha_t^2 \, |V_t| = h(G) \sum_{v \in V(G)} f(v)^2 = h(G)(f f^t)
$$

Combining this with lemma 1.22 we have

$$
h(G)(f f^t) \leq B_f \leq \sqrt{2\Delta (f L f^t)(f f^t)}
$$

which gives the following when also combined with lemma 1.21

$$
\frac{1}{2\Delta}(h(G))^2 \leq \frac{f L f^t}{f f^t} \leq \lambda
$$

This proves the lower bound. ■

We offer a polynomial time algorithm which follows from the above lower bound proof of Cheeger's inequality.

The constructive proof of the lower bound naturally provides us with a polynomial-time algorithm to find a sparse cut that approximates the Cheeger constant. By computing the second eigenvector and examining its level sets, we can find a cut that rigorously satisfies the bounds established in the preceding lemmas. This sweeping procedure is universally known as Cheeger's Algorithm.

Algorithm 1: Cheeger's Algorithm

1: Compute the eigenvector $g \in \mathbb{R}^n$ corresponding to the second smallest eigenvalue λ_2 (or λ) of the Laplacian matrix L.
2: Let $V^+ = \{v \in V : g(v) > 0\}$. If $|V^+| > \frac{|V|}{2}$, negate the eigenvector by setting $g \leftarrow -g$.
3: Construct the non-negative vector f where $f(v) = g(v)$ if $v \in V^+$, and $f(v) = 0$ otherwise.
4: Sort the distinct non-zero values of f on the vertices in strictly increasing order: $0 < \alpha_1 < \alpha_2 < \cdots < \alpha_r$.
5: **Sweep:** For each $i \in \{1, \ldots, r\}$, construct the level set $L_i = \{v \in V : f(v) \geq \alpha_i\}$.
6: Compute the edge expansion ratio for each level set: $\frac{|\partial L_i|}{|L_i|}$.

7: **Return:** The level set L^* that minimises this expansion ratio.

Because calculating eigenvectors is efficient, and there are at most n distinct level sets to sweep through and check, this entire procedure runs in polynomial time. Furthermore, the mathematical guarantees from our lower bound proof assure us that the edge expansion of the returned set L^* will strictly satisfy $\frac{|\partial L^*|}{|L^*|} \leq \sqrt{2\Delta(G)\lambda}$.

2 Metric Embeddings

2.1 Metric Spaces

In this lecture we set the stage for Bourgain's embedding theorem. We begin by recalling the definition of a metric space.

Definition 2.1. Let X be a finite set of size n. Let $d : X \times X \to \mathbb{R}$ be some function. We say that d is a ***metric on*** X and that (X, d) is a ***metric space*** if

(i) The distance function is non-negative, that is, $d(x, y) \geq 0$ for any $x, y \in X$ with the equality holds if and only if $x = y$.

(ii) The distance function is symmetric, that is, $d(x, y) = d(y, x)$ for any $x, y \in X$.

(iii) The distance function satisfies the triangle inequality. Given any $x, y, z \in X$,

$$d(x, y) + d(y, z) \geq d(x, z)$$

Below we have some examples of metrics. One may check that the above three properties are indeed satisfied.

The simplest one is the ***line metric***. Given any function $f : X \to \mathbb{R}$ we define

$$d(x, y) = |f(x) - f(y)|$$

The next one is the ***cut metric***. Given any boolean function $f : X \to \{0, 1\}$, we define

$$d(x, y) = \begin{cases} 1 & \text{if } f(x) \neq f(y) \\ 0 & \text{otherwise} \end{cases}$$

We also have the ***ℓ^p–metric*** which generalises the line metric to bigger dimensions. Given any function $f : X \to \mathbb{R}^m$ we define

$$d(x, y) = \|f(x) - f(y)\|_p$$

where if $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, then,

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$$

We shall generally prefer using it with $p = 1$ for reasons which will be clear later.

We also have the ***graph metric***. Let G be any given undirected graph on n vertices which are labelled by elements of X. We define

$$d(x, y) = d_G(x, y)$$

where the right hand side refers to the shortest distance in the graph between the two vertices x and y.

2.2 Interactions of metric spaces

We now talk about the interactions of these metrics with each other. The hierarchy is cut $\to$ line $\to$ ℓ^1 $\to$ graph.

Since we have given four examples of metrics with the graph one seeming very powerful, we need to talk about $2\binom{4}{2} = 12$ interactions. The easy chain given above takes care of 6 of these interactions as we shall soon see below.

When we traverse downwards on the chain, we lose some data in the form of distortion. This is taken care of and made precise in Bourgain's embedding theorem.

We first show that a cut metric can be thought of as a cut of a set.

Proposition 2.2. *Fix a finite set X and a subset $S \subseteq X$. Then we can induce a cut metric by choosing the indicator boolean function $1_S : X \to \{0, 1\}$. Conversely, given any cut metric, there exists a set S such that it is the cut metric with respect to this set S.*

Proof. The forward direction is clear. Conversely, let $d : X \to \mathbb{R}$ be a cut metric with respect to some boolean function $f : X \to \{0, 1\}$. Then construct $S = \{x \in X \mid f(x) = 1\}$. We claim that d is induced either by S or S^c.

Consider the metric h induced by S, that is,

$$h(x, y) = \begin{cases} 1 & \text{if } 1_S(x) \neq 1_S(y) \\ 0 & \text{if } 1_S(x) = 1_S(y) \end{cases}$$

That is, the distance, a boolean function in itself, is 1 if and only if exactly one of x or y lies in S. But $d(x, y)$ is 1 if and only if $f(x)$ and $f(y)$ are different which is the same as the above statement.

Thus, every cut metric arises from exactly two subsets of X which are complements of each other. ∎

We now start proving the three primary chain relations and list the other three as corollaries.

Proposition 2.3 (cut $\to$ line)**.** *The line metric encompasses the cut metric, that is, a cut metric is a special case of a line metric.*

Proof. Suppose we are given a cut metric d_{cut}. Obtain a set S using the above proposition 2.2. Then define the indicator function $f = 1_S : X \to \mathbb{R}$.

We claim that the line metric d_{line} arising from f is the same as d_{cut}. Now $d_{line}(x, y) = |1_S(x) - 1_S(y)|$. Since the indicator function is always 0 or 1, this output is also 0 or 1.

It is non-zero if and only if exactly one of $1_S(x)$ and $1_S(y)$ is 0 and the other is 1 which is the same as saying that exactly one of x or y lies in S while the other lies outside S, that is in the complement.

This is also the very definition of the cut metric and hence $d_{line} = d_{cut}$. ∎

Proposition 2.4 (line $\to$ ℓ^1)**.** *The ℓ^1 metric encompasses the line metric, that is, a line metric is a special case of the ℓ^1 metric.*

Proof. Suppose d_{line} is a line metric given to us. Let the associated function be f. Then choosing $m = 1$ and $g : X \to \mathbb{R}^m = \mathbb{R}$ to be $g = f$ does the job trivially. ∎

Proposition 2.5 (Any metric $\to$ graph)**.** *The graph metric encompasses every metric that is, if d is a metric then there is a graph G such that $d = d_G$*

Proof. Simply consider the complete graph on $|X|$ vertices and set the weight of the edge between vertex x and y to be $d(x, y)$. Then by the triangle inequality, this must be shortest path in the graph between x and y and hence $d_G(x, y) = d(x, y)$ and we are done. ∎

Thus on the chain, we are only left with showing that the cut metric is encompassed by the ℓ^1 metric. This readily follows as a corollary.

Corollary 2.6 (cut $\to \ell^1$)**.** *The ℓ^1 metric encompasses the cut metric, that is, a cut metric is a special case of the ℓ^1 metric.*

Proof. Combine proposition 2.3 and proposition 2.4. Explicitly, pick $m = 1$ and f to be the indicator function. ■

We now reverse the simulation direction.

We show that the cut metric encompasses the ℓ^1 metric partially. This skips two arrows in the downward direction since every line metric is an ℓ^1 metric which apparently can be expressed in terms of the cut metric. Except that it does not really skip because to show that cut metrics simulate the ℓ^1 metric, we will first need to show that they simulate line metrics.

We will only be left with the three direction of fitting the graph metric into the other three. We shall not do this as it is not possible and we will directly talk about Bourgain's embedding theorem which will cover every case since it is a general theorem about any two metrics.

Throughout, all metrics will be considered on the finite set X of cardinality $|X| = n$. Note that every metric on X can be described by a vector of size $\binom{n}{2}$ which contains information about every pair of points.

Definition 2.7. Let $\mathscr{F}$ be a family of metrics on X. Then the ***cut cone*** with respect to the family $\mathscr{F}$ is given by

$$\text{Cone}(\mathscr{F}) = \left\{ d \in \mathbb{R}^{\binom{n}{2}} \;\middle|\; d = \sum_{S \subseteq [n]} \alpha_S d_S \,;\; d_S \in \mathscr{F} \quad \alpha_S \geq 0 \right\} \setminus \{0\}$$

The motivation for the nomenclature is derived from the usual definition of a cone wherein we consider a set of generators and any positive linear combination of the generators vectors is said to lie in the cone. For instance, in $\mathbb{R}^2$, the first quadrant is a cone generated by the standard basis vectors.

Apriori, it is only known that vectors of length $\binom{n}{2}$ lie in the cone. It is not clear that every d in the cone is a metric. This is proved below.

Proposition 2.8. *Every vector in the cone of a family of metrics is a metric.*

Proof. Firstly, we do not need to verify the symmetric nature since we have only considered $\binom{n}{2}$ coordinates. We only need to verify the triangle inequality and the codomain of a metric.

The latter part is straightforward since a non-negative combination of positive vectors is also positive. Note that we had to remove the border case of the zero vector showing up.

We now verify the triangle inequality. Let a metric d (represented as a vector of length $\binom{n}{2}$) lie in the cone. Let $u, v, w \in X$. We need to show

$$d(u, w) \leq d(u, v) + d(v, w).$$

Since $d = \sum_S \alpha_S d_S$ and $\alpha_S \geq 0$, we have:

$$d(u, w) = \sum_S \alpha_S d_S(u, w) \leq \sum_S \alpha_S \big(d_S(u, v) + d_S(v, w) \big)$$

$$= \sum_S \alpha_S d_S(u, v) + \sum_S \alpha_S d_S(v, w) = d(u, v) + d(v, w).$$

■

Definition 2.9. A line metric is said to be a $[\mathbf{0}, \mathbf{1}]$*–**valued line metric*** if all the $\binom{n}{2}$ distances lie in $[0, 1]$.

Note that this is equivalent to saying that the function f defining the line metric has range $[0, 1]$.

Lemma 2.10. *Let d be a $[0,1]$–valued line metric on X. Pick t uniformly from $[0,1]$ and define a set $S_t = \{x \in X \mid f(x) > t\}$. Then for every $u, v \in X$ we have*

$$\mathbb{E}(d_{S_t}(u,v)) = d(u,v)$$

where d_{S_t} is the induced cut metric.

Proof. Suppose without loss of generality we have $f(u) \le f(v)$. Since d_{S_t} can only take boolean values, the expectation value is the same as the probability of getting a 1. Thus,

$$\begin{aligned}
\mathbb{E}(d_{S_t}(u,v)) &= P(d_{S_t}(u,v) = 1) \\
&= P(v \in S_t \text{ and } u \notin S_t) \\
&= P(f(u) \le t < f(v)) \\
&= |f(v) - f(u)| \\
&= d(u,v)
\end{aligned}$$

■

Corollary 2.11. *Every $[0,1]$–valued line metric on X can be expressed as a positive linear combination of cut metrics.*

Proof. Let $f : X \to [0,1]$ be the associated function for a $[0,1]$–valued line metric d on X. Consider the finite image of X under f in $[0,1]$ to be $0 = z_0 < z_1 < \cdots < z_k \le 1$. These are the distinct distances.

From the previous lemma 2.10 we know that

$$\begin{aligned}
d(u,v) = \mathbb{E}(d_{S_t}(u,v)) &= \int_0^1 d_{S_t}(u,v)\,dt \\
&= \sum_{i=0}^{k-1} \int_{z_i}^{z_{i+1}} d_{S_t}(u,v)\,dt \\
&= \sum_{i=0}^{k-1} d_{S_i}(u,v) \int_{z_i}^{z_{i+1}} 1\,dt
\end{aligned}$$

In the last step we observe that for a fixed $0 \le i < k$, the function d_{S_t} is constant on $t \in (z_i, z_{i+1})$. We call this constant $d_{S_i}(u,v)$ and take it out of the integral. Thus we have

$$d(u,v) = \sum_{i=0}^{k-1} \alpha_i d_{S_i}(u,v)$$

where $\alpha_i = z_{i+1} - z_i$. Now d_{S_i} is a cut metric with respect to the set $S_i = \{x \in X \mid f(x) > t\}$ for any $t \in (z_i, z_{i+1})$. This was independent of the choice of that specific t. ■

We now show that there is no need to restrict ourselves to these special kind of line metrics. We can translate and scale to obtain any line metric.

Corollary 2.12 (line → cut)**.** *Any line metric lies in the cone of the family of cut metrics.*

Proof. Let $f : X \to \mathbb{R}$ be the function from which our line metric d arises. Let $m = \min_{x \in X} f(x)$ and $M = \max_{x \in X} f(x)$. Since 0 is always in the image, $m \le 0 \le M$. Note that $M - m$ cannot be 0 else d is not a metric unless $n = 1$. We ignore such border cases.

Define $g : X \to \mathbb{R}$ given by

$$g(x) = \frac{f(x) - m}{M - m}$$

Note that g has range $[0,1]$. By the previous corollary 2.11 we know that

$$d_g(u,v) = |g(u) - g(v)| = \sum_{S \subseteq [n]} \alpha_S d_S(u,v)$$

Thus substituting for f one obtains

$$d(u,v) = |f(u) - f(v)| = (M - m)\,|g(u) - g(v)| = \sum_{S \subseteq [n]} \beta_S d_S(u,v)$$

where $\beta_S = (M-m)\alpha_S \geq 0$. This completes the proof. ■

We can now show that an ℓ^1 metric may be expressed as a conic combination of cuts.

Proposition 2.13 ($\ell^1 \to$ cut)**.** *Any ℓ^1 metric lies in the cone of the family of cut metrics.*

Proof. Let $f : X \to \mathbb{R}^m$ be the associated function of the metric d so that

$$d(u,v) = \|f(u) - f(v)\| = \sum_{i=1}^{m} |f(u)_i - f(v)_i|$$

Write $d_{g_i}(u,v) = |f(u)_i - f(v)_i|$ for every $i = 1, 2, \ldots, m$. Then one may verify that g_i is a metric and is hence a line metric corresponding to the coordinate function $f_i : X \to \mathbb{R}$. From corollary 2.12,

$$d_{g_i} = \sum_{S \subseteq [m]} \alpha_S^{(i)} d_S$$

Since $d = d_{g_1} + \cdots + d_{g_m}$, we get

$$d = \sum_{S \subseteq [m]} \left(\alpha_S^{(1)} + \cdots + \alpha_S^{(m)} \right) d_S$$

and since $\alpha_S^{(1)} + \cdots + \alpha_S^{(m)} \geq 0$ we are done. ■

Interested readers may wonder why a general ℓ^p metric fails.

We now develop theory required to state and prove Bourgain's embedding theorem.

2.3 Bourgain's Embedding Algorithm

Definition 2.14. Let $\alpha \geq 1$ be a real number. Let (X_1, d_1) and (X_2, d_2) be metric spaces. An embedding $f : X_1 \to X_2$ is said to have a ***distortion of at most α*** if there exists some $r > 0$ such that for every $x, y \in X_1$

$$r \cdot d_1(x,y) \leq d_2(f(x), f(y)) \leq r\alpha \cdot d_1(x,y)$$

Note that if X_2 is a normed vector space and the metric d_2 arises from this norm, then r may be taken to be 1. This is because we can simply consider the map $g : X_1 \to X_2$ given by $g(x) = \frac{1}{r} f(x)$ which is well defined because X_2 is a normed vector space. Since f has distortion at most alpha, we have

$$r \cdot d_1(x,y) \leq d_2(f(x), f(y)) \leq r\alpha \cdot d_1(x,y) \implies d_1(x,y) \leq d_2(g(x), g(y)) \leq \alpha \cdot d_1(x,y)$$

Thus g also has distortion α and we might as well work with g throughout.

Let us look at some examples of embedding a metric space into an ℓ^2 space.

Consider $X = \{1,2,3\}$ and $d(1,2) = d(1,3) = d(2,3) = 1$. It is indeed a metric. We can embed this easily into $(\mathbb{R}^2, \ell^2)$ by simply mapping the points to an equilateral triangle anywhere in R^2 of side length 1. The distortion is $\alpha = 1$ which is indeed the best we can do.

Consider $X = \{1, 2, 3, 4\}$ with $d(1, 2) = d(2, 3) = d(3, 4) = d(4, 1) = 1$ and $d(1, 3) = d(2, 4) = \sqrt{2}$. This is also embedded into $\mathbb{R}^2$ with distortion 1 by mapping the points to any unit square.

Note that in both the above examples, we could have embedded the metrics in R^1. In fact we can always embed any given metric in R^1 with a distortion of $12n$. This is shown in the proposition below.

Proposition 2.15. *Any metric space can be embedded in $(\mathbb{R}^1, \ell^p) = (\mathbb{R}^1, \ell^1)$ with distortion $O(n)$.*

This is a non-trivial result and we refer the reader to a paper of Matousěk. In the same paper, the author proves the bound of $12n$ while conjecturing a bound of $n - 1$.

We now outline Bourgain's embedding algorithm below.

Theorem 2.16 (Bourgain's Embedding Theorem)**.** *Any n–element metric space X can be embedded into $\mathbb{R}^{O(\log n)}$ equipped with the ℓ^1 norm with distortion $O(\log n)$.*

The proof of this theorem will require several steps - the first of which embeds the given metric space into $R^{O(\log^2 n)}$.

We describe the algorithm to do this below:

Algorithm: BourgainEmbedding(X, d)

Require: Finite metric space (X, d) with $|X| = n$
Ensure: Embedding $f : X \to \mathbb{R}^{O(\log^2 n)}$

$m \leftarrow \lceil c \log n \rceil$ $\qquad\triangleright$ c is a sufficiently large absolute constant
for $i = 1$ to $\lceil \log n \rceil$ **do**
  for $j = 1$ to m **do**
    Initialize $S_{i,j} \leftarrow \varnothing$
    for all $x \in X$ **do**
      Include x in $S_{i,j}$ independently with probability 2^{-i}
    end for
  end for
end for
for all $x \in X$ **do**
  Define
$$f(x) \leftarrow \big(d(x, S_{1,1}), d(x, S_{1,2}), \ldots, d(x, S_{\lceil \log n \rceil, m})\big)$$
end for
return f

The returned vector $f(x) \in \mathbb{R}^{m\lceil \log n \rceil}$ is the required embedding into $\mathbb{R}^{O(\log^2 n)}$. We verify that it has distortion $O(\log n)$ is the next lecture.

In the last step of the algorithm, the notation $d(x, S)$ refers to the distance of the point x from the set of points S which is the minimum of the distance from the point x to some point $s \in S$ as s varies in S.

We now wish to show the following:

$$\Omega(\log n) d(x, y) \leq \|f(x) - f(y)\|_1 \leq O(\log^2 n) d(x, y)$$

so that the distortion is indeed $O(\log n)$.

Proposition 2.17. *With respect to the algorithm we have*

$$|d(x, S) - d(y, S)| \leq d(x, y)$$

for any set $S \subseteq X$ and for every $x, y \in X$.

Proof. Without loss of generality assume that $d(y,S) \geq d(x,S)$. Suppose $x_0 \in S$ is the point closest to x so that $d(x,S) = d(x,x_0)$. Then

$$\begin{aligned}
|d(x,S) - d(y,S)| &= d(y,S) - d(x,S) \\
&= d(y,S) - d(x,x_0) \\
&\leq d(y,x_0) - d(x,x_0) \\
&\leq d(x,y)
\end{aligned}$$

■

This result proves the upper bound since $\|f(x) - f(y)\|$ is a sum of $O(\log^2 n)$ terms of the form of the left hand side of the lemma, that is, $|d(x,S) - d(y,S)|$.

Proving the lower bound is non-trivial. First off, we clarify what we mean by "balls". Given any $x, y \in X$ and radius $r \in \mathbb{R}_{>0}$ we define

$$\begin{aligned}
B(r,x) &= \{z \in X : d(x,z) \leq r\} \\
B^-(r,y) &= \{z \in X : d(y,z) < r\}
\end{aligned}$$

Proposition 2.18. *For any points $x, y \in X$, index $1 \leq i \leq \log n$ and radii $0 < r < R$, let S be some set picked by the BourgainEmbedding algorithm in the i–th iteration of the outermost loop. Suppose we have*

$$|B(x,r)| \geq 2^i\,, \ \ \left|B^-(y,R)\right| \leq 2^{i+1}\,, \ \ B(x,r) \cap B^-(y,R) = \emptyset$$

Then, with probability at least $1/64$ we have

$$R - r \leq |d(y,S) - d(x,S)|$$

Proof. Note that if the set S intersects $B_1 = B(x,r)$ and does not intersect $B_2 = B^-(y,R)$ and $B_1 \cap B_2 = \emptyset$ then indeed we must have

$$|d(y, S - d(x,S))| = d(y, S - d(x,S)) \geq R - r$$

We thus calculate the probability of S having such a property. Let $\mathcal{E}_\infty$ denote the event of S intersecting B_1 and let $\mathcal{E}_\in$ be the event of S not intersecting B_2.

Since the balls B_1 and B_2 are disjoint, the events $\mathcal{E}_1$ and $\mathcal{E}_2$ depend on the choices made by disjoint sets of vertices. Hence, they are independent events so that $\Pr(\mathcal{E}_1 \cap \mathcal{E}_2) = \Pr(\mathcal{E}_1) \cdot \Pr(\mathcal{E}_2)$.

The probability that S misses B_1 entirely is $(1-p)^{|B_1|}$. Here $p = 1/2^i$ as dictated by the algorithm. Using the inequality $1 - x \leq e^{-x}$ and the hypothesis $|B_1| \geq 2^i = 1/p$, we get:

$$\begin{aligned}
\Pr(\mathcal{E}_1) &= 1 - (1-p)^{|B_1|} \\
&\geq 1 - e^{-p \cdot |B_1|} \\
&\geq 1 - e^{-1} > \frac{1}{4}
\end{aligned}$$

Using the hypothesis $|B_2| \leq 2^{i+1} = 2/p$ and the standard estimate $(1 - \frac{1}{k})^k \approx 1/e$ (specifically $(1-2^{-i})^{2^{i+1}} \to e^{-2}$), we have for sufficiently large n:

$$\Pr(\mathcal{E}_2) = (1-p)^{|B_2|} \geq (1 - 2^{-i})^{2^{i+1}} \geq \frac{1}{16}$$

The fact that the expression is at least $1/16$ follows from straightforward results using logarithms and first term bounds for the expansion into an infinite series.

Finally,

$$\Pr(\mathcal{E}_1 \cap \mathcal{E}_2) \geq \frac{1}{4} \cdot \frac{1}{16} = \frac{1}{64}$$

■

Corollary 2.19. *Under the same assumptions as proposition 2.18, let $Z = |d(y,S) - d(x,S)|$. Then*

$$\mathbb{E}[Z] \geq \frac{1}{64}(R-r)$$

Proof. Let $\mathcal{E} = \mathcal{E}_1 \cap \mathcal{E}_2$ be the event that S intersects $B(x,r)$ and misses $B^-(y,R)$. We know from proposition 2.18 that $\Pr(\mathcal{E}) \geq 1/64$.

The random variable Z is always non-negative. Furthermore, if $\mathcal{E}$ occurs, we showed that $Z \geq R - r$. By the definition of expectation:

$$\begin{aligned}
\mathbb{E}[Z] &= \sum_v v \cdot \Pr(Z = v) \\
&\geq (R-r) \cdot \Pr(Z \geq R-r) \\
&\geq (R-r) \cdot \Pr(\mathcal{E}) \\
&\geq \frac{1}{64}(R-r)
\end{aligned}$$

■

This provides the intuition for the global lemma which we prove below.

Lemma 2.20. *Fix x, y, radii $r < R$, and index i satisfying the conditions of proposition 2.18. Let $S_{i,1}, \dots, S_{i,m}$ be the m independent sets chosen by the algorithm at scale i, where $m = c \log n$ for a large constant c.*

Let $W_i = \sum_{j=1}^{m} |d(y, S_{i,j}) - d(x, S_{i,j})|$ be the contribution of index i to the total ℓ^1 distance. Then, with probability at least $1 - n^{-3}$:

$$W_i \geq \Omega(\log n)(R-r)$$

Proof. For each $j \in \{1, \dots, m\}$, let X_j be an indicator random variable that is 1 if the event $\mathcal{E}$ (from proposition 2.18) occurs for set $S_{i,j}$, and 0 otherwise.

We know $\Pr(X_j = 1) \geq \frac{1}{64}$. Let $X = \sum_{j=1}^{m} X_j$. The expected number of "successful" sets is $\mathbb{E}[X] \geq \frac{m}{64}$. Since these trials are independent, we can apply the Chernoff bound. For $0 < \delta < 1$:

$$\Pr(X \leq (1-\delta)\mathbb{E}[X]) \leq \exp\left(-\frac{\delta^2 \mathbb{E}[X]}{2}\right)$$

Choosing $\delta = 1/2$, and recalling $m = c \log n$, we have $\mathbb{E}[X] = \frac{c \log n}{64}$.

$$\Pr(X \leq \frac{c \log n}{128}) \leq \exp\left(-\frac{c \log n}{512}\right) = n^{-c/512}$$

By choosing the constant c large enough (e.g., $c > 1536$), this probability becomes less than n^{-3}.

Thus, with high probability, at least $k = \frac{c \log n}{128}$ sets satisfy the condition. For each such set, the distance contribution is at least $R - r$. Therefore:

$$W_i \geq \sum_{j=1}^{m} X_j (R-r) \geq \frac{c \log n}{128}(R-r) = \Omega(\log n)(R-r)$$

■

Proposition 2.21. *There exists a constant C such that with probability at least $1 - \frac{1}{n}$, the embedding f satisfies*

$$\|f(x) - f(y)\|_1 \geq C(\log n) \cdot d(x,y)$$

for all $x, y \in X$ simultaneously.

Proof. Fix a pair x, y. Let $\Delta = d(x, y)$. We define a sequence of radii $t_0, t_1, \ldots, t_L$ to analyze the local growth of balls around x and y.

Set $t_0 = 0$. For $j \geq 0$, define t_{j+1} as the smallest radius $> t_j$ such that:

$$|B(x, t_{j+1})| \geq 2 \cdot |B(x, t_j)| \quad \text{OR} \quad |B(y, t_{j+1})| \geq 2 \cdot |B(y, t_j)|$$

We terminate the sequence at index L when $t_L \geq \Delta/2$. Since the maximum volume is n, the number of steps L is bounded by $2 \log n$.

Consider an interval (t_j, t_{j+1}). By definition, for any radius r in this interval, the volume of the ball around x (and y) has not yet doubled relative to t_j. Let $r = t_j$ and $R = t_{j+1}$.

We construct a valid triple for proposition 2.18. Let $p = \max(|B(x, t_j)|, |B(y, t_j)|)$. Choose the scale index i such that $2^i \leq p < 2^{i+1}$.

Without loss of generality, suppose $|B(x, t_j)|$ is the larger volume. Then:

1. $|B(x, t_j)| \geq 2^i$ (by choice of i).
2. $|B^-(y, t_{j+1})| \leq 2 \cdot |B(y, t_j)| \leq 2p < 2^{i+2}$ (since volume hasn't doubled yet).
3. $B(x, t_j) \cap B^-(y, t_{j+1}) = \emptyset$ (since $t_j + t_{j+1} \leq \Delta$).

This satisfies the conditions of proposition 2.18 (potentially using scale $i + 1$).

By lemma 2.20, with probability $1 - n^{-3}$, the contribution $W^{(j)}$ from this specific scale satisfies:

$$W^{(j)} \geq \Omega(\log n)(t_{j+1} - t_j)$$

The total distance is the sum over all scales. We sum over $j = 0$ to $L - 1$:

$$\|f(x) - f(y)\|_1 \geq \sum_{j=0}^{L-1} W^{(j)} \geq \Omega(\log n) \sum_{j=0}^{L-1} (t_{j+1} - t_j)$$

The sum telescopes to $t_L - t_0 = t_L$. Since $t_L \geq \Delta/2 = d(x, y)/2$, we have:

$$\|f(x) - f(y)\|_1 \geq \Omega(\log n) \cdot d(x, y)$$

This holds for a fixed pair x, y with probability at least $1 - n^{-3}$.

To finish, we apply the Union Bound over all pairs of vertices. There are $\binom{n}{2} < \frac{n^2}{2}$ such pairs. The probability that the condition fails for *any* pair is at most:

$$\sum_{x,y} \Pr(\text{pair } x, y \text{ fails}) < \frac{n^2}{2} \cdot n^{-3} = \frac{1}{2n} < \frac{1}{n}$$

Thus, the probability that the condition holds for *all* pairs is at least $1 - \frac{1}{n}$. ∎

So far we have shown that randomly choosing S makes the algorithm to embed succeed with probability $1 - 1/n$. This is non-zero and hence there must be some choice of the $S_{i,j}$'s for which the embedding exists.

Thus, we have proved Theorem 2.16 with dimension $O(\log^2 n)$ and distortion $O(\log n)$.

To reduce the dimension to $O(\log n)$, requires the Johnson-Lindenstrauss flattening lemma which essentially projects the set X into a space of dimension $O(\log n)$ while preserving pair-wise distances up to some small error of ε. We apply this to our vectors f sitting inside a high dimensional space and project it down to the desired $O(\log n)$ dimensional space.

This will most likely be included as one of the homework problems.

2.4 Applications of Bourgain's Theorem

We now study some applications of Bourgain's embedding theorem we proved. At this point, one may even forget that this is a spectral graph theory course.

We consider the sparsest-cut problem form graph theory. All graphs will be taken to be undirected, connected and unweighted and possibly non-simple.

Definition 2.22. Given a graph $G = (V, E)$ and a set $S \subseteq V$, we define the ***sparsity of the set*** S to be

$$SC(S) = \frac{|E(S, S^c)|}{|S|\,|S^c|}$$

The ***sparsity of the graph*** is defined to be

$$SC(G) = \min_{\emptyset \subsetneq S \subsetneq V} SC(S)$$

Observe that it is closely related to the problem of conductance, that is, definition 1.18. In fact,

$$SC(G) \leq \frac{2}{n} h(G)$$

Indeed, since $SC(S) = SC(S^c)$, we may optimise over S with size at most $n/2$. By definition we have

$$SC(G) \leq SG(S_0) = \frac{h(G)}{S_0^c} \leq \frac{2}{n} h(G)$$

where S_0 is the optimal set which yields $h(G)$ and also $h(G)$.

We now recast the sparsest cut in the language of metrics. Let 1_S be the indicator function of whether a vertex lies in S or not. Then,

$$SC(G) = \min_{\substack{S \subseteq V \\ 1 \leq |S| \leq n/2}} \frac{\sum_{u \sim v} |1_S(u) - 1_S(v)|}{\sum_{u,v} |1_S(u) - 1_S(v)|} = \min_{\substack{S \subseteq V \\ 1 \leq |S| \leq n/2}} \frac{\sum_{u \sim v} d_S(u, v)}{\sum_{u,v} d_S(u, v)}$$

where d_S is the cut metric with respect to the set S.

The problems of calculating $SC(G)$ and also obtaining the explicit sparsest cut S are both known to be NP–hard. Leighton and Rao however modelled a similar problem which can be solved in polynomial time since it is effectively reduced to a linear program.

Definition 2.23. Given a graph $G = (V, E)$ we define

$$LR(G) = \min_{d:\ d \text{ is a metric on } V} \frac{\sum_{u \sim v} d(u, v)}{\sum_{u,v} d(u, v)}$$

It follows that $LR(G) \leq SC(G)$ since d_S is a particular instance of a metric on V where S is a sparsest cut, that is, an optimising set.

Proposition 2.24. *Computing the value of $LR(G)$ takes polynomial time.*

Proof. Consider the following linear program:

$$\begin{aligned}
\text{minimise} \quad & \sum_{u \sim v \in E} d(u, v) \\
\text{subject to} \quad & \sum_{u,v} d(u, v) = 1 \\
& d(u, v) \leq d(u, w) + d(w, v) \qquad \forall\, u, v, w \in V \\
& d(u, v) \geq 0 \qquad \forall\, u, v \in V
\end{aligned}$$

We claim that the output of this program is precisely $LR(G)$. Note that scaling metrics preserves all properties of being metrics. Thus if d' is a metric which obtains the $LR(G)$ value, then scaling it by

$$\frac{1}{\sum_{u,v} d'(u,v)}$$

yields a metric d which satisfies the hypothesis of the linear program and also achieves the same $LR(G)$ value proving our claim. ∎

Theorem 2.25. *The following holds for any graph G:*

$$LR(G) \leq SC(G) \leq O(\log n) LR(G)$$

Proof. The lower bound is trivial and was also mentioned previously. We prove the upper bound. Let d^* denote an optimal metric returned by the linear program so that

$$LR(G) = \frac{\sum_{u \sim v} d^*(u,v)}{\sum_{u,v} d^*(u,v)}$$

Using Theorem 2.16, let us embed (V, d^*) into (R^k, ℓ^1), with $k = O(\log^n)$, with distortion $O(\log n)$. Let the embedding map be $f : V \to \mathbb{R}^k$. Then,

$$d^*(u,v) \leq \|f(u) - f(v)\|_1 \leq O(\log n) d^*(u,v)$$

Thus we obtain the following:

$$LR(G) = \frac{\sum_{u \sim v} d^*(u,v)}{\sum_{u,v} d^*(u,v)} \geq \frac{1}{O(\log n)} \frac{\sum_{u \sim v} \|f(u) - f(v)\|_1}{\sum_{u,v} \|f(u) - f(v)\|_1}$$

Recall that ℓ^1 is a combination of cut metrics. Thus by proposition 2.13 we have

$$\|f(u) - f(v)\|_1 = \sum_{S \subseteq [m]} \alpha_S d_S(u,v)$$

where $m = k = O(\log^2 n)$. Thus we have,

$$\begin{aligned}
LR(G) &\geq \frac{1}{O(\log n)} \frac{\sum_{u \sim v} \|f(u) - f(v)\|_1}{\sum_{u,v} \|f(u) - f(v)\|_1} \\
&= \frac{1}{O(\log n)} \frac{\sum_{u \sim v} \sum_{S \subseteq [m]} \alpha_S d_S(u,v)}{\sum_{u,v} \sum_{S \subseteq [m]} \alpha_S d_S(u,v)} \\
&= \frac{1}{O(\log n)} \frac{\sum_{S \subseteq [m]} \alpha_S \sum_{u \sim v} d_S(u,v)}{\sum_{S \subseteq [m]} \alpha_S \sum_{u,v} d_S(u,v)} \\
&\geq \frac{1}{O(\log n)} \min_{S \subseteq [m]} \frac{\sum_{u \sim v} d_S(u,v)}{\sum_{u,v} d_S(u,v)} \\
&= \frac{1}{O(\log n)} SC(G)
\end{aligned}$$

Note that the last inequality is essentially a weighted means inequalities:

$$\min_i \frac{a_i}{b_i} \leq \frac{\sum_i a_i}{\sum_i b_i} \leq \max_i \frac{a_i}{b_i}$$

Here, a_i are reals and b_i are non-zero reals all having the same sign. ∎

3 Graph Colouring Algorithms

3.1 2-Colourings and 3-Colourings

We now look at a completely different aspect of graph theory - graph colourings. We assume familiarity with basic concepts of graph colourings. As a warm up we show that colouring a bipartite graph with two colours has a polynomial time algorithm.

The graphs that will be considered throughout this colouring section shall be undirected and unweighted graphs.

Proposition 3.1. *Let G be a bipartite graph. Then there exists a polynomial time algorithm to colour it with two colours.*

Proof. The algorithm idea is straightforward. Pick a random vertex and colour it red. We colour all its neighbours blue. Given a vertex, paint all its colours with the opposite colour. We assume that the graph is connected because we could just paint component-wise.

Formally, the algorithm is clear. We need to show that it gives up the desired output which is done by showing that there are no conflicting cases. The fact that it is polynomial time is clear because we simply go over all n vertices and check if there are any uncoloured vertices left. Then we also need to go over every edge, at most twice giving the total time complexity of $O(|V| + 2\,|E|)$ which is $O(n^2)$.

We now show that the algorithm works as intended. We start with an arbitrary vertex and assign it the zeroth layer. Then we consider all its neighbours and assign them the first layer. We then consider all neighbours of this first layer that except the starting vertex and put them as the second layer.

Note that two vertices of the first layer cannot be connected else we get a triangle. We know that a graph is bipartite if and only if it has no odd cycles. We continue the process. The neighbours of layer two, excluding layer one vertices, are compiled into a third layer. Note that two third layer vertices cannot be connected as we get a 5–cycle.

In general, the $(n+1)$–th layer comprises of the neighbours of the n–th layer excluding the vertices already a part of the $(n-1)$–th layer. We observe that in our structure, edges only go between two adjacent layers and never in between the same layer. Painting the layers with alternate colours finishes the job which shows that the algorithm works. ■

There is another problem to be considered - whether a given graph is bipartite. In general, we have to decide if a graph is n–colourable and if it is, then can we find an algorithm for colouring. These are two separate problems. In the above proposition we answered the the latter question for $n = 2$ with a polynomial time algorithm.

To answer the former question for $n = 2$, we can use the same algorithm. If there is a conflicting case, which will only arise when two vertices of the same layer are adjacent, the graph fails to be bipartite. Thus, this is polynomial time, too.

The case for $n = 3$ is much harder. Deciding if a graph is 3–colourable is known to be NP-complete. Given that the graph is 3–colourable, finding a 3–colouring is also NP. The best known result is that if a graph is 3–colourable, then there is a polynomial time algorithm which produces a colouring using $O(n^{0.19747})$ colours. The exponent was initially 0.25 in 1998 when the first proof was given (KMS algorithm). Later, on there were very minor (on the surface as an exponent), although effective, improvisations. The 0.19747 exponent is by a very recent result from 2024.

Proposition 3.2. *Given that a graph is 3–colourable, there exists a polynomial time algorithm to colour it with $O\left(\dfrac{n}{\log n}\right)$ colours.*

Proof. This is Richard Lipton's naive method in which we break the vertex set into partitions of size $O(\log n)$. Since the whole graph is 3–colourable, so is each small chunk and on a given chunk of $O(\log n)$ vertices, we brute force through all the $3^{O(\log n)} = O(n)$ colourings to find a 3-colouring. We then move on to the next block, brute force with polynomial time and colour it with three entirely different colours. The process continues and the number of colours used is 3 times the number of blocks which is

$$O\Big(\frac{n}{\log n}\Big)$$

■

Avi Wigderson improved this further in a connected work.

Proposition 3.3. *Given that a graph is 3–colourable, there exists a polynomial time algorithm to colour it with $O\left(\sqrt{n}\right)$ colours.*

Proof. Recall that a graph with maximum degree Δ is colourable with $\Delta + 1$ colours in polynomial time by a naive greedy algorithm.

Let $1 \leq h \leq \Delta + 1$ be some parameter which we shall choose later. Pick any vertex v in the graph of degree at least h. Consider this vertex along with its neighbour set. If the neighbour set had chromatic number three, then we would need a fourth colour for this vertex v. This is not the case. Thus the neighbour set must be 2–colourable.

Using the polynomial time algorithm from proposition 3.1, colour this neighbour set with two colours and then give a third colour to the vertex v. Now delete this set of vertices from the graph and repeat the process on the remaining graph with a new vertex of degree at least h and give this vertex and its neighbours three new colours.

Note that at each step we throw out at least h vertices. Further, the number of steps cannot be more than n/h since there are totally n vertices. By the end of all steps, we will have a graph with all degrees being less than h and hence the remaining graph is polynomial time colourable in h colours. Thus the total number of colours used is

$$\frac{3n}{h} + h$$

Thus the total number of colours used is $O(\sqrt{n})$ for $h = \sqrt{3n}$ which minimises the expression. ■

3.2 Edge Colourings

Definition 3.4. By a k–***edge colouring*** of a simple graph G, we mean a mapping $\phi : E(G) \to [k]$ such that for any edges e and f which meet at a vertex $v = e \cap f$, $\phi(e) \neq \phi(f)$.

The ***edge chromatic number*** $\chi'(G)$ is defined to be the minimum k such that a k–edge colouring exists.

Essentially, we want to colour the edges such that ‘adjacent’ edges do not receive the same colour. This is closely related to the line graph.

Definition 3.5. Given a graph G, the ***line graph*** $L(G)$ is defined to be a graph with vertex set as the edge set $E(G)$ and two edges (new vertices) are connected if and only if they meet at a vertex in G.

It is clear that $\chi'(G) = \chi(L(G))$.

We now have some simple bounds on the edge chromatic number.

Proposition 3.6. *The edge chromatic number of a simple graph satisfies*

$$\Delta(G) \leq \chi'(G) \leq 2\Delta(G) + 1$$

Proof. The lower bound is trivial. Around the vertex with degree $\Delta(G)$, we clearly need at least $\Delta(G)$ different colours. This bound is also attained by the star graph for instance.

For the upper bound note that any graph with maximum degree $\Delta(G)$ can be coloured with $\Delta(G)+1$ colours by colouring greedily since at least colour will remain free at all times. Thus we bound $\chi'(G)$ using the maximum degree of the line graph. It is a standard fact that the maximum degree of the line graph is $2\Delta(G)-2$. The result follows. ■

The upper bound however is not achieved by any graph! This is because of the following theorem of Vizing.

Theorem 3.7 (Vizing's Theorem)**.**

$$\chi'(G) \leq \Delta(G)+1$$

Proof. The proof proceeds by induction on the number of edges. If the graph only has one edge, the maximum degree is $\Delta(G)=1$ and clearly the graph is 2–colourable. Now assume that the result holds for all graphs with $m-1$ edges. Suppose a graph has m edges.

Pick any vertex vertex x and consider its neighbours $\{y_1, \ldots, y_{d(x)}\}$. Delete the edge x—y_1. Now the remaining graph has maximum degree at most $\Delta(G)$ and is hence $\Delta(G)+1$ colourable. We assign a colour from these existing colours to the edge x—y_1 which obeys the colouring rules.

Let $\Delta(G)=\Delta$ for convenience. Let our colour palette be $\{c_1, \ldots, c_{\Delta+1}\}$. Since we have $\Delta+1$ colours and each vertex has at most Δ neighbours, the edges at each vertex must miss at least 1 colour. Suppose y_1 misses the colour c_1.

If x also misses the colour c_1, we can colour x—y_1 with c_1 and we are done. Thus suppose x does have a c_1 coloured edge. Let this edge be x—y_2 without loss of generality.

By definition of y_2, this cannot miss c_1 since y_2—x is coloured c_1. Let it miss some other colour c_2. If x also misses c_2, we can colour x—y_2 with c_2 instead of c_1 and then colour x—y_1 with c_1 and we are done.

Thu suppose x does not miss c_2 so that x—y_3 is coloured c_2. Now y_3 misses some colour which is not c_2 which is clearly present. Assume that this missing colour cannot be c_1. It has to be a new c_3.

Then if x also misses c_3, we can colour x—y_3 with c_3, which frees up c_2 so that x—y_2 can be coloured c_2 freeing up c_1 and hence x—y_1 is coloured c_1 and we are done. This argument is best seen by drawing the picture.

Thus suppose x does not miss c_3 and x—y_4 is an edge of colour c_3. We claim that y_4 cannot miss the colours c_1 and c_2 and must miss a new colour c_4. The process keeps continuing until we reach the worst case with the edge x—$y_{d(x)}$ which is coloured $c_{d(x)-1}$. Note that the process may very well end before if we y_i and x both miss c_i. Assuming that x does not miss c_i we would need to look at x—y_{i+1} and in this way we reach the worst case at the last neighbour of x.

Assume that $y_{d(x)}$ does not miss any of the colours previously seen, that is, does not miss any of $c_1, \ldots, c_{d(x)-2}$ and hence must miss a new colour $c_{d(x)}$. This is still a distinct colour in the palette since $d(x) \leq \Delta$.

Now if x also misses this new colour $c_{d(x)}$, then we can colour x—$y_{d(x)}$ with it which frees up $c_{d(x)-1}$ and so on and we are done by the shifting argument. Since the degree is x is fulfilled and we have so far managed to see the colours c_1 through $c_{d(x)-1}$, we know that x must in fact miss the colour $c_{d(x)}$ (if it did not miss the colour we need need an edge with that colour but then the degree would increase).

All that remains is to prove the following claim.

Lemma : At a given stage $i \geq 3$ suppose we have reached the edge x—y_i which is coloured c_{i-1}. Then if y_i misses any of $c_1, \ldots, c_{i-2}$, we can colour the graph and Vizing's theorem holds.

Proof of Lemma. Suppose y_i misses the colour c_j $(1 \leq j \leq i-2)$. Let x miss the colour c_0 from the palette. Delete every edge from the graph whose colour is not c_0 or c_j(including x—y_1 whose colour we do not know yet but hope to colour with c_1). What must be left is a bunch of paths and cycles (necessarily even) which must alternate in c_0 and c_j. Note that flipping the colours of the edges here will not change anything and the edge colouring still stays valid.

Now consider the vertex y_i. It must have an edge coloured c_0 (else x and y_i both miss c_0 and we can shift and finish the colouring). Continue walking along the alternating path beginning at y_i and see where it ends.

If this c_0–c_j alternating path does not end at x, we can flip the c_0 and c_j without affecting the validity of the colouring as a whole and now instead of missing c_j and having c_0, y_i now misses c_0 and has c_j. Thus we can perform the fan shifting now because x and y_i both miss c_0.

If the alternating path ends at x, it must connect to x via the c_j coloured edge since x misses c_0. But this c_j coloured edge is x—y_{j+1}! Now y_j misses a c_j coloured edge and hence cannot lie on this path. It must have its own path with the first edge from y_j being c_0 coloured. We flip everything along this edge. Since x is not on this new chain, x still misses c_0 since this chain was not affected by the flipping. Similarly y_i continues missing c_j. But y_j now misses c_0, too and we can do the fan shifting starting from y_j. \\\

■

To summarise the core mechanism of the proof, one can view the construction of the Vizing fan as a greedy algorithm. At each step, we examine the colour missed by the current neighbour y_i, say c_i, and face exactly one of three scenarios. In the first case, c_i is a new colour that is also missing at the central vertex x; here, we simply shift the colours down the fan to free up c_1 for x—y_1 and are done. In the second case, c_i is a new colour, but x does not miss it, meaning there exists an edge x—y_{i+1} coloured c_i; we are then forced to extend our fan to y_{i+1} and repeat the analysis. Because the degree $d(x) \leq \Delta$, this extension cannot continue indefinitely. Thus, we must eventually either terminate in the first case or hit the third case: the current vertex misses a colour c_j that has already appeared earlier in our sequence. This "blocked" scenario is precisely what the lemma resolves. By analysing the c_0–c_j alternating paths, we swap colours along a path to force either the current vertex or y_j to miss the same colour c_0 that x misses, thereby unblocking the fan and reducing the problem back to a simple shift.

3.3 Colour Propagation Problem

This problem generalises 2–colourings. This was proposed by Subhash Khot in 2002 and we do not know yet if this is P or NP (or perhaps both?).

Consider the following labelling problem.

Question. Suppose $k \in \mathbb{N} \setminus \{0\}$ is an integer and G is simple graph whose edges e are assigned permutations $\sigma_e \in S_k$ in a specific direction. In the other direction we use the permutation σ^{-1}. We want to check if there is a way to colour the vertices such that all permutations are satisfied. If not, we declare that the graph cannot be coloured. Can this be decided in polynomial time?

As a small example, consider the edge graph on 2 vertices with a single edge between them. Let these vertices be a, b. Suppose $k = 3$. Then we need to assign a permutation $\sigma_{(a,b)}$ in the direction from a to b. The other direction is automatically the inverse of this permutation. Thus if a has the colour red then b must necessarily have the colour σ(red).

This above idea of forcing is what we will use for our algorithm. The idea is to fix a vertex and assign it one of the colours from the list of k colours. Then force all its neighbours to have that colour. Then from the neighbours we can go to their neighbours and so on. Once all the vertices are coloured, simply check if each edge rule is satisfied.

Algorithm 2: Permutation colour Propagation

1: Iterate through each connected component of the graph G.
2: For an uncoloured vertex v in the current component, pick a colour $c \in \{1, \ldots, k\}$.
3: **Propagate:** For every neighbour w of a coloured vertex u, assign w the unique colour forced by the edge rule: $colour(w) = \sigma_{(u,w)}(colour(u)) = \sigma^{-1}_{(w,u)}(colour(u))$.
4: Repeat the propagation until all reachable vertices in the component are coloured.

5: **Verify:** Check if every edge (x, y) in the component satisfies $colour(y) = \sigma_{x,y}(colour(x))$.
6: If the verification fails, go to step 2 and attempt with the next colour for the initial vertex v.
7: If no colour in $\{1, \ldots, k\}$ results in a valid colouring, conclude the graph is not colourable.

The time complexity of the above algorithm is clearly polynomial time since we have k colour choices and we go through n vertices and check the satisfiability for at most $\binom{n}{2} = O(n^2)$ edges.

3.4 Unique Games Conjecture and Gap Problems

We make the following definition of fraction of edges satisfying the colouring assignment.

Definition 3.8. We define the ***value of a graph***, with each edge assigned a permutation with order, to be

$$\text{val}(G) = \max_{\{\ell_u\}_{u \in V}} \frac{|\{e = (u, v) : \sigma_e(\ell_u) = \ell_v\}|}{|E|}$$

Here, $\sigma \in S_k$ where k is the number of colours in the palette.

Essentially, we look at all possible colourings of vertices and consider the colouring and its corresponding largest fraction of satisfied edges.

If $val(G) = 1$, the graph is completely colourable according to the permutation rules. However, we have already established via algorithm 2 that deciding if $val(G) = 1$ is solvable in polynomial time. Therefore, conjecturing that finding the exact value of $val(G)$ is NP-hard near 1 would be trivially false.

Instead, the Unique Games Conjecture is formulated as a gap problem. We define the decision problem $UG(\epsilon, k)$ as follows.

Definition 3.9. Given a unique labelling instance, $\mathbf{UG}(\boldsymbol{\epsilon}, \mathbf{k})$ is the problem of distinguishing between the following two cases:

(i) $val(G) \geq 1 - \epsilon$ (The instance is almost entirely satisfiable).

(ii) $val(G) \leq \epsilon$ (The instance is almost entirely unsatisfiable).

Here, k is the total number of colours available in our palette.

Note that we are guaranteed somehow that one of these must necessarily occur. Choosing between them is what we are concerned with.

To understand why we formulate this as a gap problem, consider the Travelling Salesperson Problem (TSP). Asking for a factor–10 approximation of TSP is an optimisation problem. However, we can phrase it as a decision problem: given a graph, decide if the optimal cost is $\leq W$ or $\geq 10W$. If this gap decision problem is NP-hard, then finding a 10–factor approximation must also be NP-hard (since an approximation algorithm would easily distinguish between the two cases).

The Unique Games Conjecture is stated below.

Conjecture 3.10 (The Unique Games Conjecture (UGC))**.** *For every $\epsilon > 0$, there exists some $k \in \mathbb{N}$ such that $UG(\epsilon, k)$ is NP-hard.*

We shall see an algorithm of Arora, Barak and Steurer (ABS algorithm) which solves the UGC in sub-exponential time. By this, we mean that in sub-exponential time, we decide the case of $UG(\epsilon, k)$.

3.5 The ABS Algorithm and ETH

In 2010, Sanjeev Arora, Boaz Barak, and David Steurer developed an algorithm that solves the unique labelling problem in sub-exponential time. Before we delve into the spectral machinery of their approach, we must address a glaring theoretical tension this algorithm introduces.

Theorem 3.11 (Arora-Barak-Steurer)**.** *There exists an algorithm that solves the unique labelling problem $UG(\epsilon, k)$ on an N-vertex graph in sub-exponential time, specifically $2^{N^{O(\epsilon^{1/3})}}$.*

At first blush, this result can be deeply unsettling. The Unique Games Conjecture boldly asserts that $UG(\epsilon, k)$ is NP-complete for sufficiently small ϵ. If a problem is NP-complete, it is, by definition, at least as hard as 3–SAT. To understand why a sub-exponential algorithm for an NP-complete problem feels contradictory, we must formalise the Exponential Time Hypothesis.

Definition 3.12. The ***Exponential Time Hypothesis*** asserts that there exists a constant $s_3 > 0$ such that no deterministic algorithm can solve 3–SAT in time $O(2^{s_3 n})$, where n is the number of variables.

More broadly, for any fixed $k \geq 3$, the k–SAT problem requires $2^{\Omega(n)}$ time. This hypothesis is generally considered more robust and is more widely believed than the Unique Games Conjecture itself.

If we assume ETH is true, there is no $2^{o(n)}$ time algorithm for 3–SAT. Therefore, if $UG(\epsilon, k)$ is NP-complete, a sub-exponential algorithm for it seems to threaten ETH. The resolution to this apparent paradox lies in the nature of polynomial-time reductions.

It is entirely conceivable that assuming ETH does not straight away refute the Unique Games Conjecture. A polynomial-time reduction from a 3–SAT instance of size n to a Unique Games instance of size N only guarantees that $N \leq \text{poly}(n)$.

Specifically, it is possible that such a reduction produces a Unique Labelling instance of size $N = n^{O(1/\epsilon)}$. If we apply the ABS algorithm to this reduced instance, the runtime in terms of the original 3–SAT size n becomes:

$$2^{N^{O(\epsilon^{1/3})}} = 2^{(n^{O(1/\epsilon)})^{O(\epsilon^{1/3})}} = 2^{n^{O(\epsilon^{-2/3})}}$$

Since $\epsilon > 0$ is a small constant, the exponent on n can easily be ≥ 1. This means the overall runtime relative to the original 3–SAT instance is $2^{\Omega(n)}$. Thus, the ABS algorithm's sub-exponential runtime on the Unique Games instance perfectly squares with the exponential runtime lower bound demanded by ETH for 3–SAT.

Note that personally, I have no experience with the ETH and I do not understand it fully. The above short ETH interlude has been added by an acquaintance of mine.

Before examining the ABS algorithm's strategy of breaking graphs into expanders, it is instructive to consider a naive algorithmic approach. This simple strategy dictates the parameter regimes where the Unique Games Conjecture is actually non-trivial.

Proposition 3.13. *Given a unique labelling instance on a graph G with maximum degree $\Delta(G) = \Delta$, one can always efficiently find an assignment that satisfies at least $\frac{1}{\Delta+1}$ fraction of the edges.*

Proof. We can colour the edges of G such that no two adjacent edges share the same colour. By Vizing's Theorem 3.7, this requires at most $\Delta + 1$ colours. By the Pigeonhole Principle, the largest colour class (the largest set of edges sharing the same colour) must contain at least $\frac{1}{\Delta+1}$ fraction of the total edges in G.

Because no two edges in this colour class share a vertex, they form a valid matching. We can trivially satisfy all permutation constraints restricted to this matching: for each edge $e = (u, v)$ in the class, we arbitrarily assign a valid colour $\ell_u \in [k]$ to u and then assign the required colour $\ell_v = \sigma_e(\ell_u)$ to v. We can then arbitrarily assign colours to all remaining, unconstrained vertices. ■

This proposition implies that if we want to falsify the unique games conjecture for a given ϵ, we simply need $\frac{1}{\Delta+1} \geq \epsilon$. Thus, the unique labelling problem is only hard for graphs with a maximum degree at least $2/\epsilon$ (after taking a buffer; it is precisely $1/\epsilon - 1$). Moving forward, we focus our attention exclusively on d–regular graphs where $d \geq 2/\epsilon$, as this case is known to be at least as hard as the general problem.

3.6 Expander Decomposition

The high-level approach of the ABS algorithm relies heavily on the expander decomposition toolkit. The key idea is to partition the vertex set of the underlying graph into a collection of disjoint "almost expanders".

The algorithm solves the unique labelling problem on these individual pieces and gives up on the cross-edges between the pieces, ensuring that the number of such discarded edges is very small.

Definition 3.14. Let G be a graph. A partition of the vertex set into $V = V_1 \sqcup \cdots \sqcup V_k$ is called a ***δ-sparse decomposition*** if the set of cross-edges, defined as $E_{cross} = \bigcup_{i \neq j} E(V_i, V_j)$ satisfies

$$|E_{cross}| \leq \delta \, |E|$$

We now show that working with the small parts is good enough.

Theorem 3.15. *Suppose we partition V via a δ–sparse decomposition and delete the set of cross-edges E_{cross} to form a disconnected subgraph $G' = (V, E')$. If an assignment ℓ satisfies a ρ fraction of the edges in G', then ℓ satisfies at least a $\rho(1-\delta)$ fraction of the total edges in the original graph G. Furthermore, if the original instance is highly satisfiable such that $val(G) \geq 1 - \epsilon$, then the decomposed graph satisfies:*

$$val(G') \geq 1 - \frac{\epsilon}{1-\delta}$$

Proof. Let $\ell : V \to [k]$ be an assignment on the vertices. Suppose ℓ satisfies a ρ fraction of the edges in G'. The absolute number of satisfied edges in G' is precisely $\rho|E'|$.

When we apply this same assignment ℓ to the original graph G, the number of satisfied edges is at least the number of satisfied edges in G' (this assumes the absolute worst-case scenario where every single restored cross-edge in E_{cross} is violated by ℓ). Thus, the fraction of satisfied edges in G is at least:

$$\frac{\rho|E'|}{|E|} = \frac{\rho(|E| - |E_{cross}|)}{|E|} \geq \frac{\rho(|E| - \delta|E|)}{|E|} = \rho(1-\delta)$$

This proves the first claim.

For the second part, suppose $val(G) \geq 1 - \epsilon$. By Definition 3.8, this means there exists an optimal global assignment ℓ^* that violates at most $\epsilon|E|$ edges in total across the entire graph G.

In the worst-case distribution of these errors, none of these violated edges belong to E_{cross}, meaning all $\epsilon|E|$ violations occur strictly within the clusters of G'. The fraction of edges satisfied in G' by ℓ^* is therefore at least:

$$\frac{|E'| - \epsilon|E|}{|E'|} = 1 - \frac{\epsilon|E|}{|E'|}$$

Since G' was formed by a δ–sparse decomposition, we know $|E'| \geq (1-\delta)|E|$. Substituting this into the denominator gives:

$$1 - \frac{\epsilon|E|}{|E'|} \geq 1 - \frac{\epsilon|E|}{(1-\delta)|E|} = 1 - \frac{\epsilon}{1-\delta}$$

Because $val(G')$ is defined as the maximum fraction over *all* possible assignments, and we have found at least one assignment (ℓ^*) that achieves $1 - \frac{\epsilon}{1-\delta}$, it strictly follows that $val(G') \geq 1 - \frac{\epsilon}{1-\delta}$. ■

To make the notion of an "almost expander" rigorous, we define the threshold rank of a graph. Let $\epsilon > 0$ and let $\beta = O(\epsilon^{1/3})$. We also define a parameter $m = n^{O(\beta)}$.

Definition 3.16. Fix some sufficiently small $\epsilon > 0$. Let $G = (V, E)$ be a d–regular graph with lazy random walk matrix W. We define the **$(1-\epsilon)$*-threshold rank*** of G to be the number of eigenvalues of W which exceed $(1-\epsilon)$

Formally, this may be written as:

$$\mathrm{rank}_{1-\epsilon}(G) = |\{\lambda : \lambda \text{ is an eigenvalue of } W, \lambda \geq 1-\epsilon\}|$$

Recall that every eigenvalue of W is real, lies between 0 and 1 and the eigenvalue 1 is indeed achieved at the all 1's eigenvector j. Thus, the threshold rank is at least 1. The algebraic multiplicity of 1 as an eigenvalue of W is in fact the number of connected components of G.

The algorithm relies on a structural theorem that guarantees any regular graph can be partitioned into pieces with a bounded threshold rank.

Before moving to the main theorem, we tackle a related and similar problem as a warm-up.

Proposition 3.17. *Let $G = (V, E)$ be a d–regular graph on n vertices. There exists a deterministic algorithm running in polynomial time that partitions the vertex set V into disjoint subsets $V_1, V_2, \ldots, V_k$ such that the following properties hold:*

(i) For every $i \in \{1, \ldots, k\}$, the induced subgraph $G[V_i]$ is an expander with its Cheeger constant bounded below by

$$h(G[V_i]) \geq \frac{d}{\log^3 n}$$

(ii) The total number of edges crossing the partition is strictly sub-linear in the number of vertices, meaning

$$\left| \bigcup_{1 \leq i < j \leq k} E(V_i, V_j) \right| = o(n)$$

Proof. We first present a structural, non-algorithmic charging argument to obtain this partitioning, which can then be adapted into a polynomial-time algorithm using spectral techniques.

Let $h^* = \frac{d}{\log^3 n}$ be our target Cheeger constant. If the initial graph satisfies $h(G) \geq h^*$, the graph is already the desired expander, and we terminate the process without removing any edges.

Otherwise, there exists a sparse cut (V_S, V_S^c) such that $|V_S| \leq |V|/2$ and the edge expansion is strictly less than h^*. This implies the number of edges crossing the cut is bounded by:

$$|\partial V_S| < h^* |V_S|$$

We remove the edges crossing this cut to split the graph into two disjoint components, V_S and V_S^c. To ensure the components remain exactly d–regular for subsequent steps, for every deleted edge $e = (u, v)$, we add a self-loop at u and a self-loop at v within their respective new components. We can visualise this process as building a binary tree level by level, where the root is V and the children are the partitioned components.

We recursively apply this procedure to any component A at the leaves of our current tree. If the induced graph satisfies $h(G[A]) \geq h^*$, we stop and mark the component as a finalised expander. If not, we find a cut (A_S, A_S^c) with $|A_S| \leq |A|/2$ and expansion less than h^*, remove the crossing edges (at most $h^* |A_S|$), add self-loops, and recurse.

To bound the total number of edges removed, we employ a charging argument. At any intermediate step where a component A is split, let A_S be the smaller side ($|A_S| \leq |A|/2$). We collect h^* "money" from every vertex in the smaller side A_S. The total money collected from these vertices is exactly $h^* |A_S|$, which strictly upper-bounds the number of edges lost in this specific refinement step, since $|\partial A_S| < h^* |A_S|$.

Whenever a vertex pays this cost, it is placed into a component that is at most half the size of its previous component. Consequently, throughout the entire recursive process, a single vertex can pay at most $\log_2 n$ times before it ends up in an isolated component of size 1.

The total amount paid by a single vertex is therefore bounded by $h^* \log_2 n$. Summing over all n vertices, the absolute maximum number of edges lost across the entire decomposition is:

$$\text{Total edges cut} \leq n \cdot h^* \log_2 n = n \cdot \left(\frac{d}{\log^3 n} \right) \log_2 n = \frac{dn \log_2 n}{\log^3 n}$$

Since $\frac{dn \log_2 n}{\log^3 n}$ scales as $O\left(\frac{dn}{\log^2 n} \right)$, the total number of edges removed across the entire graph is strictly $o(n)$. At the end of this process, every resulting component V_i is a guaranteed expander with $h(G[V_i]) \geq d/\log^3 n$.

To make this process algorithmic and run in polynomial time, we can use Cheeger's algorithm 1 to recursively find the sparse cuts. Recall that Cheeger's inequality guarantees that if a cut of expansion h^* exists, sweeping the eigenvector will find a cut with expansion at most $\sqrt{2\Delta h^*}$. In our case this is

$$\sqrt{2\Delta h^*} = \sqrt{2d\left(\frac{d}{\log^3 n}\right)} = O\left(\frac{d}{\log^{1.5} n}\right)$$

Thus, instead of charging each vertex $h^* = d/(\log^3 n)$ amount of money, we now charge it $d/(\log^{1.5} n)$ amount of money. Hence the total amount paid by all n vertices would be

$$n \cdot \log n \cdot O\left(\frac{d}{\log^{1.5} n}\right) = O\left(\frac{dn}{\sqrt{\log n}}\right)$$

Note that this also continues to be in the class of $o(n)$.

Thus, we have converted the theoretical cuts into algorithmic cuts and although the algorithmic ones allow more expansion, we still remain in the $o(n)$ class. ■

We can now deal with and state the main theorem of this section on expanders.

Theorem 3.18. *The vertex set of every d–regular n–vertex graph can be partitioned into disjoint subsets $V_1, V_2, \ldots, V_b$ such that each subset induces a graph which has $\mathrm{rank}_{1-\epsilon}(G[V_i]) \leq m = n^{O(\beta)}$ for all i. Further, the number of edges between these disjoint sets is at most 1% of the total number of edges in G.*

The proof of this theorem is constructive and relies on recursively finding and removing sparse cuts. If a piece of the graph is not yet an "almost expander" (meaning its threshold rank is too high), we call this an offending piece. The following critical lemma ensures that whenever we have an offending piece, we can efficiently find a small, non-expanding set to cut out.

Lemma 3.19. *Fix $\epsilon > 0$. There exists an algorithm which, given a d–regular graph H with $n_H \leq n$ vertices and $\mathrm{rank}_{1-\epsilon}(H) \geq m = n^{O(\beta)}$ (i.e., $\lambda_m(W_H) \geq 1 - \epsilon$), runs in $\mathrm{poly}(n)$ time and returns a set $S \subseteq V(H)$ satisfying:*

(i) $|S| \leq \frac{n_H}{m}$

(ii) The edge expansion of the set S is bounded by

$$\frac{|\partial S|}{|S|} \leq O\Big(d\sqrt{\frac{\epsilon}{\beta}}\Big)$$

Using this lemma it is now easy to prove Theorem 3.18. We provide the proof below.

Proof. (of Theorem 3.18) The algorithm from the lemma recursively partitions the graph to eliminate any "offending pieces" (induced subgraphs with a threshold rank strictly greater than m). We begin with the entire graph G.

At any stage, if an induced subgraph H on n_H vertices is an offending piece, meaning $\mathrm{rank}_{1-\epsilon}(H) > m$, we apply lemma 3.19. The lemma guarantees we can efficiently find a cut (S, S^c) within $V(H)$ such that:

$$|S| \leq \frac{n_H}{m} \quad \text{and} \quad |\partial S| \leq C \cdot d\sqrt{\frac{\epsilon}{\beta}}\,|S|$$

where C is the hidden absolute constant from the Big-O notation.

We remove the crossing edges ∂S, split H into two disjoint subgraphs induced by S and S^c, and add self-loops to all affected vertices to ensure both new components remain exactly d–regular. We then recursively repeat this process on any resulting component that still has a threshold rank greater than m.

We use the charging-type argument from before. Every time a component H is split, we distribute the "cost" of the cut edges strictly among the vertices in the smaller set S. At most $Cd\sqrt{\frac{\epsilon}{\beta}}\,|S|$ edges are removed implying that each vertex of S was charged exactly $Cd\sqrt{\frac{\epsilon}{\beta}}$.

Whenever a vertex pays this charge, it is placed into a new component S whose size is at most $\frac{1}{m}$ times the size of its previous component H. Thus, the size of the component containing this vertex drops by a factor of at least m. A single vertex can undergo this size reduction at most $\log_m n$ times before it ends up in an isolated component of size 1, at which point it can never be cut again.

Recall our parameter $m = n^{c\beta}$ (where c is the hidden constant in the exponent). This gives us:

$$\log_m n = \frac{\log n}{\log(n^{c\beta})} = \frac{1}{c\beta}$$

Thus, a single vertex is charged at most $\frac{1}{c\beta}$ times. The absolute maximum cost paid by any single vertex across the entire recursive process is:

$$\text{Max charge per vertex} \leq \frac{1}{c\beta} \cdot \left(Cd\sqrt{\frac{\epsilon}{\beta}}\right) = \frac{C}{c} \cdot d \cdot \frac{\sqrt{\epsilon}}{\beta^{3/2}}$$

Recall that we set $\beta = \Theta(\epsilon^{1/3})$. Therefore, $\beta^{3/2} = \Theta(\sqrt{\epsilon})$, which means the ratio $\frac{\sqrt{\epsilon}}{\beta^{3/2}}$ is bounded by an absolute constant.

By choosing the constant c (the scaling factor in the exponent of m) to be sufficiently large, we can make the fraction $\frac{C}{c}$ arbitrarily small. Specifically, we can choose c such that the maximum charge per vertex is strictly less than $0.01\frac{d}{2}$.

Summing this cost over all n vertices, the total number of removed edges across the entire decomposition is bounded by:

$$\text{Total edges cut} \leq n \cdot \left(0.01\frac{d}{2}\right) = 0.01\left(\frac{dn}{2}\right)$$

Since $\frac{dn}{2}$ is exactly the total number of edges in the original d–regular graph G, the number of edges crossing the disjoint sets is at most 1% of the total edges, completing the proof. ■

The difficult part is to prove the lemma. To prove this lemma, the ABS algorithm introduces a size-aware variant of the Cheeger bound. Standard Cheeger rounding might yield a sparse cut, but it does not guarantee the support of the cut is small. To overcome this, we look for vectors that are not just sparse in the boolean sense, but sparse in the real-valued sense.

Definition 3.20. A real-valued vector $x \in \mathbb{R}^n$ is called δ–***analytically sparse*** if its 1–norm and 2–norm satisfy the following inequality:

$$\|x\|_1 \leq \sqrt{\delta n} \cdot \|x\|_2$$

If the random walk fails to mix well after several steps (which is guaranteed by the high threshold rank), the resulting probability distribution vector will have a large 2–norm, making it analytically sparse. One can then perform a truncation and rounding procedure on this analytically sparse vector to extract the desired small-sized set S with low conductance.

Lemma 3.21. *Fix $\epsilon > 0$ and let H be a regular graph on n_H vertices. Suppose the $(1-\epsilon)$ threshold rank of H is bounded below by $m = n^{O(\beta)}$, meaning $\lambda_m(W_H) \geq 1-\epsilon$. Let $t = \frac{\beta}{\epsilon}\log n$. Then, there exists a vertex $u \in V(H)$ such that the random walk distribution after t steps satisfies:*

$$\left\|W_H^t \mathbf{1}_u\right\|_2^2 \geq \frac{n^\beta}{n_H}$$

Proof. For notational convenience, let $W = W_H$. We analyse the trace of the matrix W^{2t}. By definition, the trace is the sum of the eigenvalues. Since $\lambda_1 \geq \cdots \geq \lambda_m \geq 1 - \epsilon$, we can bound the trace from below by looking only at the top m eigenvalues:

$$\mathrm{Tr}(W^{2t}) = \sum_{i=1}^{n_H} \lambda_i(W^{2t}) \geq \sum_{i=1}^{m} \lambda_i(W)^{2t} \geq m(1-\epsilon)^{2t}$$

Substituting $t = \frac{\beta}{\epsilon} \log n$ and using the standard approximation $(1-\epsilon)^{1/\epsilon} \approx e^{-1} \geq 2^{-1.5}$, we get:

$$m(1-\epsilon)^{2t} = n^{O(\beta)} \left((1-\epsilon)^{1/\epsilon} \right)^{2\beta \log n} \geq n^{O(\beta)} n^{-\beta} \geq n^{\beta}$$

The last inequality holds by choosing the hidden constant in $O(\beta)$ to be sufficiently large.

Alternatively, the trace can be computed by summing the diagonal entries of W^{2t}. Let $\mathbf{1}_u$ be the indicator vector for vertex u. Then:

$$\mathrm{Tr}(W^{2t}) = \sum_{u \in V(H)} \mathbf{1}_u^T W^{2t} \mathbf{1}_u = \sum_{u \in V(H)} \mathbf{1}_u^T W^t W^t \mathbf{1}_u = \sum_{u \in V(H)} \left\| W^t \mathbf{1}_u \right\|_2^2$$

Since the sum of these n_H positive terms is at least n^{β}, an averaging argument guarantees there must exist at least one vertex $u \in V(H)$ such that the term $\|W^t \mathbf{1}_u\|_2^2$ is at least the average value:

$$\left\| W^t \mathbf{1}_u \right\|_2^2 \geq \frac{n^{\beta}}{n_H}$$

■

Lemma 3.22. *Let H be a d–regular graph on n_H vertices with lazy random walk matrix $W = W_H$. Suppose $x \in \mathbb{R}^{n_H}$ is a δ–analytically sparse vector which satisfies $\frac{\|Wx\|_2^2}{\|x\|_2^2} \geq (1-\alpha)$ for some $\alpha > 0$.*

Then, there exists a set $S \subseteq V(H)$ with cardinality $|S| \leq 4\delta n_H$ such that the edge expansion is bounded by:

$$\frac{|\partial S|}{|S|} \leq O\left(d\sqrt{\alpha}\right)$$

Proof. First, we relate the 2–norm condition to the Rayleigh quotient. Since W is symmetric and $W - W^2$ is positive semi-definite (meaning correlation drops as random walks evolve), we have:

$$\frac{x^T W x}{\|x\|_2^2} \geq \frac{x^T W^2 x}{\|x\|_2^2} = \frac{\|Wx\|_2^2}{\|x\|_2^2} \geq 1 - \alpha$$

Recall that for a d–regular graph, the lazy random walk matrix is $W = \frac{1}{2}\left(I + \frac{1}{d}A\right) = I - \frac{1}{2d}L$, where $L = dI - A$ is the standard Laplacian matrix. Substituting this into our inequality gives:

$$\frac{x^T \left(I - \frac{1}{2d}L\right) x}{x^T x} \geq 1 - \alpha \implies 1 - \frac{1}{2d}\frac{x^T L x}{x^T x} \geq 1 - \alpha \implies \frac{x^T L x}{x^T x} \leq 2d\alpha$$

Because x is δ–analytically sparse, we can scale it such that $\|x\|_2^2 = \delta n_H$, which simultaneously forces $\|x\|_1 \leq \delta n_H$. We construct a strictly non-negative, truncated vector $y \in \mathbb{R}^{n_H}$ defined point-wise as:

$$y_u = \begin{cases} x_u - 1/4 & \text{if } x_u \geq 1/4 \\ 0 & \text{otherwise} \end{cases}$$

Since y_u is strictly non-zero only when $x_u \geq 1/4$, and the absolute sum of all components is bounded by $\|x\|_1 \leq \delta n_H$, the number of non-zero entries in y can be at most $\frac{\delta n_H}{1/4} = 4\delta n_H$. Thus, the support of y satisfies $|\mathrm{supp}(y)| \leq 4\delta n_H$.

We now evaluate the Laplacian Rayleigh quotient for y. Since truncation cannot increase the difference between any two values, $(y_i - y_j)^2 \le (x_i - x_j)^2$, meaning the numerator $y^T L y \le x^T L x$. For the denominator, since $y_u \ge 0$, we have $y_u^2 \ge x_u^2 - x_u/2$. Summing over all vertices:

$$y^T y \ge x^T x - \frac{1}{2}\|x\|_1 \ge \delta n_H - \frac{1}{2}\delta n_H = \frac{1}{2}\|x\|_2^2$$

Combining these, the Rayleigh quotient of y is bounded by:

$$\frac{y^T L y}{y^T y} \le \frac{x^T L x}{\frac{1}{2} x^T x} \le 2(2d\alpha) = 4d\alpha$$

Finally, we apply Cheeger's Algorithm to this vector y. By our earlier constructive proof of the lower bound (proposition 3.17), sweeping the level sets of y guarantees the existence of a cut S such that:

$$\frac{|\partial S|}{|S|} \le \sqrt{2\Delta(G)\left(\frac{y^T L y}{y^T y}\right)} \le \sqrt{2d(4d\alpha)} = \sqrt{8d^2\alpha} = O\left(d\sqrt{\alpha}\right)$$

Furthermore, since Cheeger's algorithm selects a level set from the non-zero entries of y, we have $S \subseteq \mathrm{supp}(y)$, guaranteeing $|S| \le 4\delta n_H$. ■

We can now finally prove lemma 3.19.

Proof. (of lemma 3.19) For ease of notation, let $W = W_H$. From lemma 3.21, we know there exists a vertex $u \in V(H)$ such that after $t = \frac{\beta}{\epsilon}\log n$ steps, the random walk fails to mix, yielding:

$$\left\|W^t \mathbf{1}_u\right\|_2^2 \ge \frac{n^\beta}{n_H} \ge \frac{n^\beta}{n}$$

We claim there must exist a pair of successive steps, s and $s+1$ (where $0 \le s < t$), such that the 2–norm does not drop significantly:

$$\left\|W^{s+1}\mathbf{1}_u\right\|_2^2 \ge \left(1 - \frac{\epsilon}{\beta}\right)\left\|W^s \mathbf{1}_u\right\|_2^2$$

If this condition failed for every single step from 0 to t, we would have $\|W^t \mathbf{1}_u\|_2^2 \le \left(1 - \frac{\epsilon}{\beta}\right)^t \|\mathbf{1}_u\|_2^2$. Since $\|\mathbf{1}_u\|_2^2 = 1$ and $(1 - \frac{\epsilon}{\beta})^t \approx e^{-t\epsilon/\beta} = e^{-\log n} = \frac{1}{n}$, this would imply $\|W^t \mathbf{1}_u\|_2^2 \ll \frac{1}{n}$, which directly contradicts our lower bound. Thus, such a step s must exist.

Let $x = W^s \mathbf{1}_u$. Since x is a probability distribution vector resulting from a random walk, its entries sum to 1 and are non-negative, meaning $\|x\|_1 = 1$. The failure of the 2–norm to drop implies Wx has a large 2–norm:

$$\|Wx\|_2^2 \ge \left(1 - \frac{\epsilon}{\beta}\right)\|x\|_2^2$$

Furthermore, because $\|Wx\|_2 \le \|x\|_2$ for any lazy random walk matrix, we also have $\|x\|_2^2 \ge \|Wx\|_2^2 \ge \left(1 - \frac{\epsilon}{\beta}\right)\frac{n^\beta}{n} \ge \frac{1}{\delta n_H}$ for $\delta = \frac{1}{m}$ (since $m = n^{O(\beta)}$). Because $\|x\|_1 = 1$ and $\|x\|_2 \ge \frac{1}{\sqrt{\delta n_H}}$, x strictly satisfies the definition of being δ–analytically sparse.

Since x is δ–analytically sparse and satisfies $\frac{\|Wx\|_2^2}{\|x\|_2^2} \ge 1 - \frac{\epsilon}{\beta}$, we can directly invoke lemma 3.22 with $\alpha = \frac{\epsilon}{\beta}$ and $\delta = \frac{1}{m}$. The lemma guarantees the existence of a set $S \subseteq V(H)$ such that $|S| \le 4\frac{n_H}{m} = O\left(\frac{n_H}{m}\right)$ and:

$$\frac{|\partial S|}{|S|} \le O\left(d\sqrt{\frac{\epsilon}{\beta}}\right)$$

Absorbing the constant 4 into the asymptotic notation of $m = n^{O(\beta)}$ gives $|S| \le \frac{n_H}{m}$, concluding the proof. ■

3.7 Solving UGC on Almost Expanders

So far in our discussion, we did not use anything specific to the Unique Labelling Problem. Indeed, the preceding section only partitioned the underlying graph into almost expanders while completely ignoring the permutation labels! In this section, we present the ideas which underlie the second step of the ABS algorithm, showing how one can find a good unique games labelling on an almost expanding instance. We first introduce a helpful auxiliary graph.

For the rest of this section, we assume G is an almost expander (a low-rank graph) produced by our decomposition. We also perform a convenient reparametrisation by setting $\epsilon := \gamma^6$. Let us formally collect all the parameters we will use going forward:

- $n \in \mathbb{N}$: The number of vertices in the original graph G.
- $0 < \gamma \ll 0.01$.
- $\epsilon = \gamma^6$: The threshold parameter.
- $\beta = O(\epsilon^{1/3}) = O(\gamma^2)$.
- $m = n^{O(\beta)}$: The threshold rank bound.

Definition 3.23. Let $UG(G, \alpha, k)$ denote an instance of the unique labelling problem where $G = (V, E)$ is the underlying graph. We define the ***Label-Extended Graph*** $\hat{G} = (\hat{V}, \hat{E})$ as follows.

The vertices of $\hat{G}$ have the form (v, i) where $v \in V(G)$ and $i \in [k]$. This is to say, for each $v \in V(G)$, we define its ***cloud*** as:

$$\text{CLOUD}(v) = \{(v,1), (v,2), \ldots, (v,k)\}$$

which contains exactly k copies of the vertex v, one in each colour. Note that the total number of vertices is $\left|\hat{V}\right| = kn$.

An edge in $\hat{G}$ is of the form $((u, a), (v, b))$ where $u, v \in V(G)$ and the given edge rule π_{uv} satisfies $\pi_{uv}(a) = b$. In other words, for every edge $(u, v) \in E(G)$, we include a perfect matching of k edges in $\hat{E}$ between $\text{CLOUD}(u)$ and $\text{CLOUD}(v)$ that satisfies the edge rule π_{uv} in the natural way.

Suppose the unique labelling instance is completely satisfiable, and the satisfying assignment assigns the colour $\sigma(v)$ to each $v \in V(G)$. This means the specific set $S \subseteq \hat{V}$ defined by $S = \{(v, \sigma(v))\}_{v \in V(G)}$ is completely disconnected from the rest of $\hat{V}$. This occurs because any edge leaving S would imply a violation of the permutation constraints, which contradicts complete satisfiability. The real utility of the Label-Extended Graph stems from a robust version of this observation: a highly satisfiable unique labelling instance corresponds to a sparse cut in $\hat{G}$.

Lemma 3.24. *Consider an instance $UG(G, \alpha, k)$ of the unique labelling problem where $G = (V, E)$ is a d–regular graph with $d \geq 2/\alpha$. Suppose this instance admits an assignment that satisfies at least a $(1 - \alpha)$ fraction of the edges. Let $\hat{G}$ denote the label-extended graph of G. Then, there exists a set $S \subseteq \hat{V}$ such that its edge expansion is bounded by:*

$$\frac{|\partial S|}{|S|} \leq 2\alpha d$$

Proof. Note that the label-extended graph $\hat{G}$ is also exactly d–regular, because every vertex in $\text{CLOUD}(u)$ has exactly one edge going to $\text{CLOUD}(v)$ for each neighbour v of u in G.

Denote by $\sigma : V(G) \to [k]$ the assignment that satisfies the maximum fraction of the edges of G. Consider the set of vertices corresponding to this assignment in the label-extended graph:

$$S = \{(v, \sigma(v))\}_{v \in V(G)} \subseteq \hat{V}$$

Notice that the cardinality is exactly $|S| = n$. We wish to bound the edge expansion of this set S.

Take an edge $(u, v) \in E(G)$ that is violated by σ (meaning $\pi_{uv}(\sigma(u)) \neq \sigma(v)$). We can pair this violated edge in G with exactly two edges in $\hat{G}$ that are incident to the vertices $(u, \sigma(u))$ and $(v, \sigma(v))$ in S. Because the edge is violated, these two edges do not connect to each other within S, and must therefore cross the cut (S, S^c) in $\hat{G}$.

Since we know that at most an α fraction of the total $nd/2$ edges in G are violated, the total number of cut edges in $\hat{G}$ is bounded by twice this amount:

$$|\partial S| \leq 2 \cdot \alpha \left(\frac{nd}{2}\right) = \alpha dn$$

Substituting $n = |S|$, the edge expansion of S is bounded by:

$$\frac{|\partial S|}{|S|} \leq \frac{\alpha d\,|S|}{|S|} = \alpha d \leq 2\alpha d$$

This proves the existence of a cut with the desired sparsity. ■

At this stage, our mathematical intuition might reflexively ask if the converse to lemma 3.24 holds. That is, might a sparse cut in the label-extended graph automatically correspond to a good labelling for our unique labelling instance?

Unfortunately, this is not necessarily true. The sparsest cut might correspond to some set $S \subseteq \hat{V}$ which contains more than one vertex from the same cloud, or misses some clouds entirely. Sometimes, it is still useful to proceed undeterred. Kolla-Tulsiani and ABS show that we can pull out our spectral toolkit and analyse the span of eigenvectors corresponding to large eigenvalues of the lazy random walk matrix $\hat{W}$ on the label-extended graph $\hat{G}$. The key is to show this space contains a real-valued vector which can be rounded into a sparse cut that in turn corresponds to a good assignment, provided G is an almost expander.

To solidify this approach, we must answer a few key questions:

(i) The span of "top" eigenvectors of $\hat{W}$ only contains a good vector which corresponds to a sparse cut. How do we efficiently find it?

(ii) How exactly do we extract a valid assignment from this sparse cut?

(iii) We only know G is an almost expander. We want to analyse the top eigenspace of $\hat{W}$ (not W). This requires a rigorous bound on the dimensionality of this top eigenspace. How do we obtain that?

Before proceeding to answer these questions, we make a structural observation that reaffirms the spectral usefulness of label-extended graphs. Consider an instance $UG(G, \alpha, k)$ of the unique labelling problem where $G = (V, E)$ is a d–regular graph with $d \geq 2/\alpha$. Let $\hat{G}$ denote its label-extended graph. We observe that corresponding to every cycle in G, there are exactly k corresponding cycles in $\hat{G}$.

A bound on the number of cycles in $\hat{G}$ allows us to bound the trace of some appropriate power of the lazy random walk matrix $\hat{W}$ on $\hat{G}$. The trace is the sum of the eigenvalues of $\hat{W}$ raised to that power, which provides the crucial spectral connection we will exploit to answer the third question from our list.

3.8 The Top Eigenspace $\hat{W}$

We begin by formally defining the eigenspace we wish to analyse. We want to search for our sparse cut indicator vector within the span of the top eigenvectors of the label-extended graph's random walk matrix.

Definition 3.25. Let $\hat{W}$ denote the lazy random walk matrix for the label-extended graph $\hat{G}$. The $(1-\gamma^5)$–***top eigenspace*** of $\hat{W}$, denoted as $\mathcal{U}$, is defined as the span of all eigenvectors whose corresponding eigenvalues are at least $1 - \gamma^5$:

$$\mathcal{U} = \text{SPAN}\left(\left\{x : x \text{ is an eigenvector of } \hat{W} \text{ with eigenvalue } \geq 1 - \gamma^5\right\}\right)$$

To efficiently brute-force a search through this subspace, we must rigorously guarantee that its dimension, $\dim(\mathcal{U})$, is strictly bounded by $m = n^{O(\beta)}$. We achieve this through a sequence of spectral trace bounds.

First, we formalise a powerful spectral property of the almost expanders produced by our earlier decomposition. Because an almost expander has very few eigenvalues near 1, the trace of its random walk matrix raised to a sufficiently high power remains small.

Lemma 3.26. *Let $G[V_i]$ be an almost expander produced by the termination of our expander decomposition algorithm (Theorem 3.18), and let W be its lazy random walk matrix. For the random walk depth $t = \frac{\beta}{\epsilon}\log n$, the trace is strictly bounded by:*

$$\operatorname{Tr}(W^{2t}) < n^{\beta}$$

Proof. We prove this not through algebraic eigenvalue bounds, but by the operational guarantees of the decomposition algorithm itself.

Recall the contrapositive of the logic used to find an offending piece (lemma 3.21). During the decomposition, we established that if a graph satisfies $\operatorname{Tr}(W^{2t}) \geq n^{\beta}$, an averaging argument guarantees the existence of a vertex u where $\|W^t \mathbf{1}_u\|_2^2 \geq \frac{n^{\beta}}{n_H}$. This analytically sparse vector is then used to find a sparse cut, strictly forcing the graph to be partitioned further.

Therefore, if $G[V_i]$ is a finalised component returned by the algorithm, it means the graph cannot be partitioned any further by our subroutine. The algorithm's inability to find an offending piece mathematically necessitates that the precondition for the cut failed. Thus, it must strictly be true that:

$$\operatorname{Tr}(W^{2t}) < n^{\beta}$$

Since $\beta = O(\epsilon^{1/3})$, we have $n^{\beta} \leq n^{O(\beta)}$, which provides our desired sub-exponential bound. ■

We now show that this trace bound lifts beautifully to the label-extended graph $\hat{G}$. By observing the combinatorial relationship between closed walks in G and $\hat{G}$, we bypass the need for ad-hoc algebraic rank definitions.

Lemma 3.27. *Let W and $\hat{W}$ be the lazy random walk matrices for G and $\hat{G}$ respectively. For any integer $t \geq 1$, we have:*

$$\operatorname{Tr}(\hat{W}^{2t}) \leq k \operatorname{Tr}(W^{2t})$$

Proof. The trace of an adjacency matrix raised to the power $2t$ counts the exact number of closed walks of length $2t$. For the lazy random walk matrix $W = \frac{1}{2}\left(I + \frac{1}{d}A\right)$, the trace $\operatorname{Tr}(W^{2t})$ is directly proportional to the number of valid closed lazy walks.

In the label-extended graph $\hat{G}$, a walk of length $2t$ corresponds to a walk in the base graph G alongside a tracking of the colour permutation. A walk in $\hat{G}$ is closed if and only if its projection in G is closed AND the composed permutation along the walk maps the starting colour exactly back to itself.

For any given closed walk in G, there are k possible starting colours in $\hat{G}$. Since the permutation constraints along the edges are deterministic, each of these k starting colours traces out exactly one valid walk in $\hat{G}$. Thus, a single closed walk in G lifts to at most k closed walks in $\hat{G}$ (it achieves exactly k if the net permutation is the identity, and strictly fewer otherwise).

Because both graphs are d–regular, this combinatorial bound on the number of closed walks transfers directly to the trace of their respective random walk matrices, yielding $\operatorname{Tr}(\hat{W}^{2t}) \leq k \operatorname{Tr}(W^{2t})$. ■

With these two trace bounds established, we can now elegantly prove the main dimensionality theorem, completely resolving the open exercise regarding the eigenspace of $\hat{W}$.

Theorem 3.28. *Let G be an almost expander. The dimension of the $(1 - \gamma^5)$–top eigenspace of the label-extended graph $\hat{W}$ is strictly sub-exponential. Specifically,* $\dim(\mathcal{U}) \leq n^{O(\beta)}$.

Proof. Let $R = \dim(\mathcal{U})$ be the number of eigenvalues of $\hat{W}$ that are at least $1-\gamma^5$. By the definition of the trace, we can bound $\mathrm{Tr}(\hat{W}^{2t})$ from below by restricting the sum to just these R top eigenvalues:

$$\mathrm{Tr}(\hat{W}^{2t}) = \sum_{i=1}^{kn} \hat{\lambda}_i^{2t} \geq \sum_{i=1}^{R} \hat{\lambda}_i^{2t} \geq R\left(1-\gamma^5\right)^{2t}$$

From lemma 3.26 and lemma 3.27, we have the upper bound $\mathrm{Tr}(\hat{W}^{2t}) \leq k\,\mathrm{Tr}(W^{2t}) \leq k(m+1)$. Combining these inequalities gives:

$$R\left(1-\gamma^5\right)^{2t} \leq k(m+1) \implies R \leq k(m+1)\left(1-\gamma^5\right)^{-2t}$$

Recall we chose the random walk depth $t = \frac{\beta}{\epsilon}\log n$. Using the standard approximation $(1-x)^{-1} \approx e^x$ for small x, we can bound the exponential penalty term:

$$\left(1-\gamma^5\right)^{-2t} \leq \exp\left(2t\gamma^5\right) = \exp\left(2\frac{\beta}{\epsilon}(\log n)\gamma^5\right)$$

Substituting our parameters $\epsilon = \gamma^6$ and $\beta = c\gamma^2$ (for some hidden scaling constant c), the exponent simplifies:

$$2\frac{c\gamma^2}{\gamma^6}\gamma^5 \log n = 2c\gamma \log n$$

Thus, the exponential penalty is bounded by $\exp(2c\gamma\log n) = n^{2c\gamma} = n^{O(\gamma)}$. Because $\gamma = \epsilon^{1/6}$ and $\beta = O(\epsilon^{1/3})$, we mathematically have $\gamma = o(\beta)$, meaning $n^{O(\gamma)}$ is strictly dominated by $n^{O(\beta)}$.

Substituting this penalty back into our bound for R, and recalling $m = n^{O(\beta)}$ while k is a constant:

$$R \leq k \cdot n^{O(\beta)} \cdot n^{O(\gamma)} = n^{O(\beta)}$$

This proves that the dimensionality of the top eigenspace $\mathcal{U}$ is bounded by $n^{O(\beta)}$, justifying our ability to efficiently enumerate it. ∎

3.9 Finding a Good Assignment in the Top Eigenspace

We have successfully bounded the dimensionality of the $(1-\gamma^5)$–top eigenspace, $\mathcal{U}$, ensuring that a brute-force search over an appropriate net is computationally feasible in sub-exponential time. We must now guarantee that this search will actually yield a useful vector.

Recall from lemma 3.24 that an optimal assignment satisfying at least a $(1-\epsilon)$ fraction of the edges corresponds to a set $S \subseteq \hat{V}$ with edge expansion at most $2\epsilon d$. We define the normalised indicator vector of this set as $\chi_S = \frac{\mathbf{1}_S}{\sqrt{|S|}}$. The following lemma demonstrates that the projection of χ_S onto our top eigenspace $\mathcal{U}$ captures nearly all of its L_2 mass.

Lemma 3.29 (Indicator Projection). *Let $P_{\mathcal{U}}$ denote the orthogonal projection operator onto the subspace $\mathcal{U}$. Then the squared norm of the projected vector satisfies:*

$$\|P_{\mathcal{U}}\chi_S\|_2^2 \geq 0.99\,\|\chi_S\|_2^2$$

Proof. For any unit vector $z \in \mathcal{U}$, its eigenvalue with respect to the lazy random walk matrix $\hat{W}$ is at least $1-\gamma^5$, meaning $z^T\hat{W}z \geq 1-\gamma^5$.

From our edge expansion bound on S, the Rayleigh quotient of the indicator vector with respect to the Laplacian is $\frac{\chi_S^T\hat{L}\chi_S}{\chi_S^T\chi_S} \leq 2\epsilon d$. Since $\hat{W} = I - \frac{1}{2d}\hat{L}$, we have:

$$\chi_S^T\hat{W}\chi_S = \|\chi_S\|_2^2 - \frac{1}{2d}\chi_S^T\hat{L}\chi_S \geq \|\chi_S\|_2^2 - \frac{2\epsilon d}{2d}\|\chi_S\|_2^2 = (1-2\epsilon)\|\chi_S\|_2^2$$

Let $\alpha = \|P_{\mathcal{U}}\chi_S\|_2^2$. By the Orthogonal Decomposition Theorem, we can uniquely write $\chi_S = x + y$, where $x \in \mathcal{U}$ and $y \in \mathcal{U}^{\perp}$. Consequently, $\|x\|_2^2 = \alpha$ and $\|y\|_2^2 = \|\chi_S\|_2^2 - \alpha$.

We now bound the quadratic form $\chi_S^T \hat{W} \chi_S$ from above. Since x and y are orthogonal, the cross-terms vanish:

$$\chi_S^T \hat{W} \chi_S = x^T \hat{W} x + y^T \hat{W} y$$

The term $x^T \hat{W} x$ is bounded by the maximum eigenvalue of $\hat{W}$, which is 1, giving $x^T \hat{W} x \le \|x\|_2^2 = \alpha$. The vector y lies entirely in the orthogonal complement $\mathcal{U}^{\perp}$, meaning it is spanned only by eigenvectors with eigenvalues strictly less than $1 - \gamma^5$. Thus, $y^T \hat{W} y \le (1 - \gamma^5)\|y\|_2^2$. This yields the upper bound:

$$\chi_S^T \hat{W} \chi_S \le \alpha + \left(\|\chi_S\|_2^2 - \alpha\right)(1 - \gamma^5)$$

Since $\epsilon = \gamma^6$, we know that $\gamma^5 = \epsilon^{5/6}$. For sufficiently small ϵ, we easily have $\gamma^5 \ge 1000\epsilon$. Substituting this relaxes the bound to:

$$\chi_S^T \hat{W} \chi_S \le \alpha + \left(\|\chi_S\|_2^2 - \alpha\right)(1 - 1000\epsilon)$$

Chaining our lower and upper bounds together, we obtain:

$$(1 - 2\epsilon)\|\chi_S\|_2^2 \le \alpha + \left(\|\chi_S\|_2^2 - \alpha\right)(1 - 1000\epsilon)$$

Expanding the right side gives $\alpha + \|\chi_S\|_2^2 - 1000\epsilon\|\chi_S\|_2^2 - \alpha + 1000\epsilon\alpha$. Simplifying and rearranging terms:

$$998\epsilon\|\chi_S\|_2^2 \le 1000\epsilon\alpha \implies \alpha \ge \frac{998}{1000}\|\chi_S\|_2^2 > 0.99\|\chi_S\|_2^2$$

This confirms that $\|P_{\mathcal{U}}\chi_S\|_2^2 \ge 0.99\|\chi_S\|_2^2$, as desired. ■

This lemma guarantees that the top eigenspace $\mathcal{U}$ contains a vector remarkably close to the ideal sparse cut indicator. To isolate it, the ABS algorithm employs a brute-force subspace enumeration over a carefully constructed net.

Algorithm 3: Subspace Enumeration

1: **Input:** A label-extended graph $\hat{G}$ and the gap parameter ϵ.
2: Let $\mathcal{U} = \text{SPAN}\left\{x : x \text{ is an eigenvector of } \hat{W} \text{ with eigenvalue } \ge 1 - \gamma^5\right\}$.
3: Let $m = \dim(\mathcal{U})$ and initialize an empty collection of candidate sets $\mathcal{C} = \emptyset$.
4: Construct Σ_m, a $\left(0.1\sqrt{\gamma}\right)$–net covering the m–dimensional unit ball of $\mathcal{U}$.
5: **For** each vector $v \in \Sigma_m$:
6: **For** each threshold index $s \in \{1, 2, \ldots, n\}$:
7: Define the level set: $S_{v,s} = \left\{i \in \hat{V} : v(i) \ge \frac{1}{2\sqrt{s}}\right\}$.
8: Add $S_{v,s}$ to the collection $\mathcal{C}$.
9: **Return** the collection $\mathcal{C}$.

Because $\dim(\mathcal{U}) = m = n^{O(\beta)}$, the size of the net Σ_m is bounded by $\exp(n^{O(\beta)})$, keeping the overall runtime strictly sub-exponential. The enumeration guarantees that the returned collection $\mathcal{C}$ contains at least one set $\tilde{S} \subseteq \hat{V}$ such that its edge expansion is $O(\epsilon)$ and its symmetric difference with the optimal assignment set S is exceedingly small: $\left|S \oplus \tilde{S}\right| \le O(\gamma\,|S|)$.

We now arrive at the final step: extracting a valid colouring from this approximation set $\tilde{S}$. We will rigorously prove the validity of this assignment, showing precisely how the small symmetric difference guarantees a near-optimal solution.

Definition 3.30. Given a candidate set $\tilde{S} \subseteq \hat{V}$ from algorithm 3, we define the ***Extracted Assignment*** $\tilde{\sigma} : V(G) \to [k]$ for the underlying graph G as follows:

(i) If the intersection $\tilde{S} \cap \mathrm{CLOUD}(v)$ is non-empty, assign $\tilde{\sigma}(v)$ to be an arbitrary colour present in this intersection.

(ii) If $\tilde{S}$ entirely misses $\mathrm{CLOUD}(v)$, assign $\tilde{\sigma}(v)$ an arbitrary colour from $[k]$.

Theorem 3.31. *Let $S = \{(v, \sigma^*(v))\}$ be the set corresponding to the optimal unique games assignment σ^* that satisfies a $(1-\epsilon)$ fraction of the edges in G. If $\left|S \oplus \tilde{S}\right| \le O(\gamma\,|S|)$, then the extracted assignment $\tilde{\sigma}$ satisfies at least a $(1 - O(\gamma))$ fraction of the edges in G.*

Proof. Recall that the underlying graph G is d–regular with n vertices and exactly $\frac{nd}{2}$ edges. The optimal assignment set S has exactly one vertex per cloud, meaning $|S| = n$. The optimal assignment σ^* violates at most $\epsilon\frac{nd}{2}$ edges.

We analyse the number of vertices where our extracted assignment $\tilde{\sigma}$ differs from the optimal assignment σ^*. For a vertex $v \in V(G)$ to have $\tilde{\sigma}(v) \neq \sigma^*(v)$, the correct vertex $(v, \sigma^*(v))$ must have either been omitted from $\tilde{S}$, or an incorrect vertex (v, c) must have been included in $\tilde{S}$ and subsequently chosen by our arbitrary tie-breaking rule.

In both scenarios, such a vertex directly contributes at least 1 to the symmetric difference $\left|S \oplus \tilde{S}\right|$. Let $B \subseteq V(G)$ denote the set of these "bad" vertices. We can strictly bound their number:

$$|B| \le \left|S \oplus \tilde{S}\right| \le O(\gamma\,|S|) = O(\gamma n)$$

Now, we count the number of edges in G that might be violated by our extracted assignment $\tilde{\sigma}$. An edge $e = (u, v)$ can only be violated if it falls into one of two categories:

(i) The edge was already violated by the optimal assignment σ^*. There are at most $\epsilon\frac{nd}{2}$ such edges.

(ii) The edge was satisfied by σ^*, but at least one of its endpoints lies in the bad set B.

Since G is d–regular, the maximum number of edges incident to the vertices in B is exactly $d\,|B|$. Therefore, the total number of newly violated edges is bounded by:

$$d\,|B| \le d \cdot O(\gamma n) = O(\gamma n d)$$

The total number of edges violated by $\tilde{\sigma}$ is at most the sum of the originally violated edges and the newly corrupted edges:

$$\text{Total Violated} \le \epsilon\frac{nd}{2} + O(\gamma n d)$$

Dividing by the total number of edges $\frac{nd}{2}$ to find the fraction of violated edges yields:

$$\text{Fraction Violated} \le \epsilon + O(\gamma)$$

Because $\epsilon = \gamma^6$, the ϵ term is strictly dominated by $O(\gamma)$. Thus, the fraction of violated edges is bounded by $O(\gamma)$. Consequently, the extracted assignment $\tilde{\sigma}$ successfully satisfies at least a $1 - O(\gamma)$ fraction of all edges in G.

Since $\gamma \lll 0.01$ is a small constant, this efficiently recovered assignment is an excellent approximation, completing the Arora-Barak-Steurer sub-exponential algorithm for the Unique Games problem. ■

4 Lovász-Simonovits Curve Technique

4.1 Introduction and Motivation

We circle back to random walks on graphs in this section. We saw that using the second smallest eigenvalue and corresponding eigenvector of the Laplacian we could find a "sparse" cut. But to do this we needed to look at the full matrix of the graph globally and we need a global view to get the eigenvector and eigenvalue of the Laplacian.

To do this faster, it would be nice if we could do things locally and find this sparse set quicker without a lot of error. Indeed, the concept of random walks comes into play naturally. We start our walker in this sparsely connected small set S and let it "walk" randomly. It has a high chance of staying within S and a low chance of going outside since S is sparsely connected to the rest of the graph.

4.2 Edge Distribution and Lazy Walks

Recall that we preferred working with the lazy random walk matrix defined in definition 1.11. In the graph world, for every vertex v of degree d_v, we added d_v many loops at the vertex and took the adjacency matrix of this new graph which was called the lazy random walk matrix. Clearly the number of edges doubled.

We make another minor modification, the reason for which will become clearer as the theory proceeds. We treat every edge in our undirected graph as two directed edges. Thus, we have begun with our graph having $|E|$ number of undirected edges and ended up with a directed graph with $4|E|$ number of directed edges. Since the directed edges occur in pairs, the matrix is still the lazy random walk matrix and is still symmetric.

As seen before we considered the distribution p on the vertices and saw how fast $\|pW^t - \pi\|$ shrunk where π is the stationary distribution given by

$$\pi(v) = \frac{d_v}{2|E|}$$

This is from Theorem 1.14.

We now shift our focus to edge distributions. Consider a vertex v and all of its outgoing edges. Initially it has d_v many undirected edges and now it has $2d_u$ many outgoing (and an equal number of incoming) edges. For each of these edges, we assign the weight

$$\frac{p(v)}{2d_v}$$

The total weight of all $4d_u$ directed edges is still 1. Here, p is some probability distribution on the vertices.

As $p \to \pi$ when we iterate W, each edge weight tends to

$$\frac{1}{4|E|}$$

That is, every edge will have the same weight and clearly all weights still add up to 1.

The idea now is to track the entire edge weight probability distribution rather than a single number (like the L_2 distance between p and π as considered in the random walks section).

4.3 The Lovász-Simonovits Greedy Potential

Definition 4.1. Let ρ be the edge probability mass distribution induced by a vertex distribution p. Sort all the $4|E| = m$ edges in decreasing order of mass as

$$\rho(e_1) \geq \cdots \geq \rho(e_m)$$

We define the *discrete Lovász-Simonovits potential* $G_p : \{0, 1, \ldots, m\} \to [0, 1]$ as follows:

$$G_p(x) = \sum_{i=1}^{x} \rho(e_i)$$

We define the ***Lovász-Simonovits potential*** $g_p : [0, m] \to [0, 1]$ to be the linear interpolation of G_p (interpolated between successive integers).

Note that the function is well defined since the output value is always positive since each edge weight is positive and since the sum of all edge weights is 1. It is also clear that $G_p(x) = g_p(x)$ is 0 at $x = 0$ and 1 at $x = m$. Further, both of these are increasing functions since we are taking partial sums of non-negative terms. There is another extremely important observation we state as a proposition below. This will be used throughout the lecture

Proposition 4.2. *The function g_p is concave for any probability distribution p on the vertices.*

Proof. Since the curve between integers of $[0, m]$ is just a straight line, for any point $x \in (k-1, k) \subseteq [0, m]$, the slope of this line segment is given by the slope of the line joining $(k-1, G_p(k-1))$ and $(k, G_p(k))$. Here we are considering the slope of the k-th line segment and hence $1 \leq k \leq m$. Thus,

$$g'(x) = \frac{G_p(k) - G_p(k-1)}{k - (k-1)} = \rho(e_k)$$

Thus it is clear that the slope $g'(x)$ is a non-increasing function which proves that g is concave. Note that we cannot test for the derivative at these integers since the function is piecewise linear and clearly not differentiable at these points. We only need that the left hand derivative at each point is at least the right hand derivative which is indeed the case here. ∎

We make another observation that for the distribution $p = \pi$, the function g_p is simply

$$g_p(x) = \frac{x}{m}$$

This is clear since the partial sums keep adding the constant $1/m$ at every point and interpolating just gives us the big straight line.

Proposition 4.3. *The curve g_π always lies below g_p for every p, that is $g_p(x) \geq g_\pi(x)$ for every $x \in [0, m]$.*

Proof. One may choose to go completely algebraically and prove this computationally. It suffices to show the claim at integers x since both the curves are interpolations. However, the concavity proposition is really nice and we can use it directly. Since the curve is concave, any line segment joining two points on it must lie below the curve. Consider the two points to be $(0, 0)$ and $(m, 1)$. The line segment joining these two points is exactly the image of the curve g_π and we are done. ∎

Now we ask that if $p \to \pi$ then does $g_p \to g_\pi$ pointwise? The answer is yes, at least by intuition. The remainder of this section proves this fact.

Let us say LS instead of Lovász-Simonovits. Further, let us begin with the probability distribution p_0 concentrated at some vertex v_0 (that is 1 at coordinate v_0 and 0 elsewhere). Let the iterates be $p_t = p_0 W^t$ where W is the lazy random walk matrix. Let g_{p_t} be shortened to g_t for all $t \geq 0$.

We first show that with each iteration the new curves do not rise above the former curves. We need to develop some theory before doing so. The result is found in corollary 4.6

Definition 4.4. Let p be a distribution on V (vertex set). Arrange the edges with respect to their induced weights ρ as

$$\rho(e_1) \geq \cdots \geq \rho(e_m)$$

Observe that for a given vertex, every edge weight is the same since its probability is equally distributed among its $2d_u$ out edges. Thus, with respect to the above ordering, a point $x \in \{0, 1, 2, \ldots, m\}$ is called a ***hinge point*** if the x heaviest edges considered up to this point correspond to *all* the out-edges of some subset of vertices of V.

Essentially, for every vertex v, consider the ratio $\frac{p(v)}{2d_v}$. Order the vertices as $v_1 \succeq \cdots \succeq v_n$ according to this ratio from highest to lowest. Then the first $2d_{v_1}$ values $\rho(e_1), \ldots, \rho(e_{2d_{v_1}})$ all correspond to the out edges of v_1 and so on. Thus the hinge points are $0, 2d_{v_1}, 2d_{v_1} + 2d_{v_2}, \ldots, m$. Note that $m = 2d_{v_1} + \cdots + 2d_{v_n} = 4\,|E|$.

Proposition 4.5. *Fix $t \geq 1$. Then for every hinge point of g_t, we have $g_t(x) \leq g_{t-1}(x)$.*

Proof. Let x be a hinge point. Then the x largest edges are all the out edges corresponding to some set of vertices $S \subseteq V$. Thus the greedy potential may be written as

$$
\begin{aligned}
g_t(x) &= \sum_{u \in S} \sum_{e \in E_{\text{out}}(u)} \rho_t(e) \\
&= \sum_{u \in S} 2d_u \frac{p_t(u)}{2d_u} \\
&= \sum_{u \in S} p_t(u)
\end{aligned}
$$

Now recall that by the iterative construction we have $p_t = p_{t-1} W$ so that

$$
\begin{aligned}
p_t(u) &= \sum_{v \in N(u)} p_{t-1}(v) W_{v,u} \\
&= \sum_{v \in N(u)} \frac{p_{t-1}(v)}{2d_v}
\end{aligned}
$$

Now observe that the summand is simply the edge weight of the edge from v to u. Thus, we have

$$
g_t(x) = \sum_{u \in S} \sum_{e \text{ to u}} \rho_{t-1}(e)
$$

where "e to u" refers to edges which originate somewhere but their heads are at u, that is, of the form $e = (z, u)$ for some $z \in V$. Corresponding to every such vertex u we get $2d_u$ number of in edges. Suppose they are $f_1^{(u)}, \ldots, f_{2d_u}^{(u)}$. Then we get

$$
g_t(x) = \sum_{u \in S} \sum_{j=1}^{2d_u} \rho_{t-1}(f_j^{(u)})
$$

The total number of terms being added here is $\sum_{u \in S} 2d_u = x$. For each of these x terms we are adding the edge weight ρ_{t-1} corresponding to some edges. But we know that our greedy function is truly greedy and $g_{t-1}(x)$ is the sum of the largest x values of ρ_{t-1} and hence we get

$$
g_t(x) \leq g_{t-1}(x)
$$

Note as an additional remark that the equality would hold if and only if these x edges $f_j^{(u)}$ are precisely the largest x edges by weight in some order. ∎

Corollary 4.6. *For every $t \geq 1$ and every $x \in [0, m]$, we have*

$$
g_t(x) \leq g_{t-1}(x)
$$

Proof. By proposition 4.5 we know that the new curve drops below the old curve at the hinge points. Let $x \in [0, 2m]$ be some real number. Assume it is not a hinge point else we are done. Suppose it lies between the consecutive hinge points $x_a < x < x_b$. Since $[x_a, x_b] \subseteq \mathbb{R}$ is convex let us write

$$
x = \lambda x_a + (1 - \lambda) x_b
$$

for some $\lambda \in (0, 1)$. Between the hinge points x_a and x_b, every vertex contributes the same constant value to the partial sum and hence g_t is a straight line and hence

$$
g_t(x) = \lambda g_t(x_a) + (1 - \lambda) g_t(x_b) \leq \lambda g_{t-1}(x_a) + (1 - \lambda) g_{t-1}(x_b)
$$

Note that the inequality followed from the previous proposition (we know the newer curve is at most older curve at hinge points x_a and x_b). However, we cannot apply linearity of g_{t-1} here because x_a and x_b may not be hinge points of g_{t-1}. However we use concavity from proposition 4.2. Thus the line segment joining $(x_a, g_{t-1}(x_a))$ and $(x_b, g_{t-1}(x_b))$ must lie below the curve $g_{t-1}(x)$ and hence

$$g_t(x) \leq \lambda g_{t-1}(x_a) + (1-\lambda) g_{t-1}(x_b) \leq g_{t-1}(\lambda x_a + (1-\lambda) x_b) = g_{t-1}(x)$$

■

We interpret $g_t(x)$ as a solution to some natural linear program. The LS curve aggregates edges greedily by heaviness with respect to ρ_t. Taking inspiration from the continuous (fractional) knapsack problem, and writing c_{e_i} for the i-th component of a vector $\vec{c}$, we get exactly:

$$\begin{aligned} g_t(x) = \text{maximise} \quad & \vec{c} \cdot \vec{\rho_t} \\ \text{subject to} \quad & 0 \leq c_{e_i} \leq 1 \quad \forall\ 1 \leq i \leq m \\ & \sum_{i=1}^{m} c_{e_i} = x \end{aligned}$$

Here the vectors lie in $\mathbb{R}^m$ where $m = 4\,|E|$.

For those not entirely comfortable with viewing this through the lens of linear programming, we can present this result in a more elementary, algebraic way.

Proposition 4.7. *For every sequence of real numbers $c_1, c_2, \ldots, c_m$ such that $0 \leq c_i \leq 1$, we have*

$$\sum_{i=1}^{m} c_i \rho_t(e_i) \leq g_t\left(\sum_{i=1}^{m} c_i\right)$$

Proof. Let $x = \sum_{i=1}^{m} c_i$. We are looking to bound the sum $\sum_{i=1}^{m} c_i \rho_t(e_i)$ given the constraint that each coefficient $c_i \in [0, 1]$.

Recall the construction of the Lovász-Simonovits potential. By definition, $g_t(x)$ evaluates the maximum possible mass one can accumulate by taking exactly x "units" of edges. Because the sequence of edges is sorted in non-increasing order of weight ($\rho_t(e_1) \geq \rho_t(e_2) \geq \ldots$), the optimal greedy strategy to maximise the accumulated weight is to take the full weight ($c_i = 1$) of the heaviest edges first, until we reach x.

If x is an integer, $g_t(x)$ is exactly the sum of the first x edges. If x is not an integer, the linear interpolation means $g_t(x)$ takes the first $\lfloor x \rfloor$ edges fully, and takes a fractional amount $(x - \lfloor x \rfloor)$ of the next heaviest edge.

Since the vector $\vec{c} = (c_1, \ldots, c_m)$ simply represents *some* arbitrary fractional allocation of exactly x units across the edges, the total weight yielded by this allocation can never exceed the optimal greedy allocation represented by $g_t(x)$. Thus, the inequality holds immediately. ■

4.4 Deflation Recurrence and Converge Rates

We now establish a recurrence that bounds how much the Lovász-Simonovits curve drops at each step, assuming the graph has a certain conductance. We briefly recall what conductance is (so far we have refrained from using conductance and worked with the Cheeger constant $h(G)$).

Definition 4.8. Given an undirected graph $G = (V, E)$ and a subset $S \subseteq V$, we define the volume of S to be the sum of the degrees of its vertices, $vol(S) = \sum_{u \in S} d_u$. The ***conductance of the set*** S is given by:

$$\varphi(S) = \frac{|\partial S|}{vol(S)}$$

The ***conductance of the graph*** G, denoted $\varphi(G)$, is defined as the minimum conductance over all subsets that contain at most half the total volume of the graph:

$$\varphi(G) = \min_{\substack{S \subseteq V \\ 0 < vol(S) \leq \frac{vol(V)}{2}}} \frac{|\partial S|}{vol(S)}$$

Observe that for a d-regular graph, $vol(S) = d\,|S|$. In this uniform case, the conductance is simply a scaled version of the Cheeger constant, yielding the direct relation $\varphi(G) = \frac{1}{d}h(G)$.

Theorem 4.9. *Let $G = (V, E)$ be the graph defined before and suppose the conductance of our graph is $\varphi(G) = \phi$. For every start distribution p_0, for every $t \geq 1$, and for every hinge point x of g_t:*

- *If $x \leq m/2$, then*
$$g_t(x) \leq \frac{g_{t-1}(x - 2\phi x) + g_{t-1}(x + 2\phi x)}{2}$$
- *If $x > m/2$, then*
$$g_t(x) \leq \frac{g_{t-1}(x - 2\phi(m - x)) + g_{t-1}(x + 2\phi(m - x))}{2}$$

Proof. We prove the case for $x \leq m/2$. The case for $x > m/2$ follows analogously. Let x denote a hinge point of g_t corresponding to a subset of vertices $S \subseteq V$. Thus, $\sum_{u \in S} 2d_u = x$. From the proof of proposition 4.5, we know that $g_t(x)$ is exactly the mass that entered S at step $t-1$:

$$g_t(x) = \sum_{f \in E_{\text{in}}(S)} \rho_{t-1}(f)$$

We partition $E_{\text{in}}(S)$ into two disjoint sets:

- $W_1 = \{(v, u) : u, v \in S, u \neq v\}$ (non-loop edges fully inside S)
- $W_2 = \{(v, u) : u \in S, v \notin S\} \cup \{(u, u) : u \in S\}$ (edges entering S from outside, plus all loops on S)

We define two new sets of edges to utilise the greedy property of the potential function. Form W_1' by taking each edge $e = (v, u) \in W_1$ and pairing it with the unique loop at v. Form W_2' by pairing each non-loop edge $e \in W_2$ with a unique loop at its source, and pairing each loop $e \in W_2$ with a unique out-edge from its vertex.

Since each pair has equal probability mass under ρ_{t-1}, we have

$$\sum_{e \in W_1} \rho_{t-1}(e) = \frac{1}{2} \sum_{e \in W_1'} \rho_{t-1}(e) \leq \frac{1}{2} g_{t-1}(|W_1'|)$$
$$\sum_{e \in W_2} \rho_{t-1}(e) = \frac{1}{2} \sum_{e \in W_2'} \rho_{t-1}(e) \leq \frac{1}{2} g_{t-1}(|W_2'|)$$

Let E_{in} be the number of undirected internal edges in S, and $C = \left|E(S, \overline{S})\right|$ be the number of undirected crossing edges. By the handshaking lemma, the volume of the set is $vol(S) = 2E_{\text{in}} + C$. In our directed lazy graph, the hinge point x equals the total out-degree of S, giving $x = 2vol(S) = 4E_{\text{in}} + 2C$.

By construction, $|W_1| = 2E_{\text{in}}$ and $|W_2| = C + vol(S) = 2E_{\text{in}} + 2C$. Thus, the sizes of our paired sets are:

$$|W_1'| = 2\,|W_1| = 4E_{\text{in}} = x - 2C$$
$$|W_2'| = 2\,|W_2| = 4E_{\text{in}} + 4C = x + 2C$$

Note that their average is exactly x. Since the graph is a ϕ-expander, the conductance bounds the crossing edges: $\phi \leq \frac{C}{vol(S)}$. Since $vol(S) = x/2$, this implies $2C \geq \phi x$. Therefore, $|W_1'| \leq x - \phi x$ and $|W_2'| \geq x + \phi x$.

Summing our two bounds together and applying the concavity of g_{t-1} from proposition 4.2, we obtain:

$$\begin{aligned} g_t(x) &\le \frac{g_{t-1}(|W_1'|) + g_{t-1}(|W_2'|)}{2} \\ &\le \frac{g_{t-1}(x - \phi x) + g_{t-1}(x + \phi x)}{2} \end{aligned}$$

where the concavity guarantees that symmetrically expanding the points outward from $[|W_1'|, |W_2'|]$ to $[x - \phi x, x + \phi x]$ around the centre x strictly lowers the chord midpoint. ■

We finally come to the end of the section and state and prove the main theorem on convergence of the g_t to g_π.

Theorem 4.10. *Suppose the conductance of a graph G is at least ϕ. Then for every start distribution p_0, every $t \ge 1$, and every $x \in [0, m]$, we have*

$$g_t(x) \le \min\left\{\sqrt{x}, \sqrt{m - x}\right\} \left(1 - \frac{\phi^2}{8}\right)^t + \frac{x}{m}$$

Proof. Let us define a family of reference curves $R_t(x)$ representing the right-hand side of our inequality:

$$R_t(x) = \min\left\{\sqrt{x}, \sqrt{m - x}\right\} \left(1 - \frac{\phi^2}{8}\right)^t + \frac{x}{m}$$

We proceed by induction on t. For the base case $t = 0$, we must verify that $g_0(x) \le R_0(x)$ for all $x \in [0, m]$. Since p_0 is a point distribution, the maximum mass $g_0(x)$ can accumulate from any subset of edges is trivially bounded by 1, and for small x, it is bounded by x. Since $x \le \sqrt{x}$ on $[0, 1]$ and $1 \le \sqrt{x}$ for $x \ge 1$, the bound $g_0(x) \le \sqrt{x}$ holds. By the symmetry of the domain, $g_0(x) \le R_0(x)$ is satisfied everywhere.

Assume the bound holds for $t - 1$, such that $g_{t-1}(x) \le R_{t-1}(x)$ for all $x \in [0, m]$. We wish to show $g_t(x) \le R_t(x)$. Because both g_t and R_t are concave functions, and g_t is piecewise linear between its hinge points, it suffices to prove the inequality exclusively at the hinge points of g_t.

Let x be a hinge point of g_t. By symmetry, we may assume $x \le m/2$. Applying the deflation recurrence from Theorem 4.9 and our inductive hypothesis, we obtain:

$$\begin{aligned} g_t(x) &\le \frac{g_{t-1}(x - \phi x) + g_{t-1}(x + \phi x)}{2} \\ &\le \frac{R_{t-1}(x - \phi x) + R_{t-1}(x + \phi x)}{2} \end{aligned}$$

We now substitute the definition of R_{t-1} into the right-hand side. We evaluate the minimums by using the mathematical fact that $\min\{A, B\} \le A$ and $\min\{A, B\} \le B$. For the first term, we bound the minimum by its first argument: $\min\left\{\sqrt{x - \phi x}, \sqrt{m - (x - \phi x)}\right\} \le \sqrt{x - \phi x}$. For the second term, we also bound the minimum by its first argument: $\min\left\{\sqrt{x + \phi x}, \sqrt{m - (x + \phi x)}\right\} \le \sqrt{x + \phi x}$. Applying these relaxed bounds yields:

$$\begin{aligned} g_t(x) &\le \frac{1}{2}\left(\sqrt{x - \phi x}\left(1 - \frac{\phi^2}{8}\right)^{t-1} + \frac{x - \phi x}{m} + \sqrt{x + \phi x}\left(1 - \frac{\phi^2}{8}\right)^{t-1} + \frac{x + \phi x}{m}\right) \\ &= \frac{1}{2}\left(\sqrt{x - \phi x} + \sqrt{x + \phi x}\right)\left(1 - \frac{\phi^2}{8}\right)^{t-1} + \frac{2x}{2m} \\ &= \frac{1}{2}\sqrt{x}\left(\sqrt{1 - \phi} + \sqrt{1 + \phi}\right)\left(1 - \frac{\phi^2}{8}\right)^{t-1} + \frac{x}{m} \end{aligned}$$

By the Taylor expansion bounds for $\phi \in (0,1)$, we have the inequality $\frac{1}{2}\left(\sqrt{1-\phi}+\sqrt{1+\phi}\right) \leq 1 - \frac{\phi^2}{8}$. Substituting this yields:

$$\begin{aligned} g_t(x) &\leq \sqrt{x}\left(1 - \frac{\phi^2}{8}\right)\left(1 - \frac{\phi^2}{8}\right)^{t-1} + \frac{x}{m} \\ &= \sqrt{x}\left(1 - \frac{\phi^2}{8}\right)^{t} + \frac{x}{m} \\ &= R_t(x) \end{aligned}$$

This completes the induction, proving the theorem for all $t \geq 1$. ■